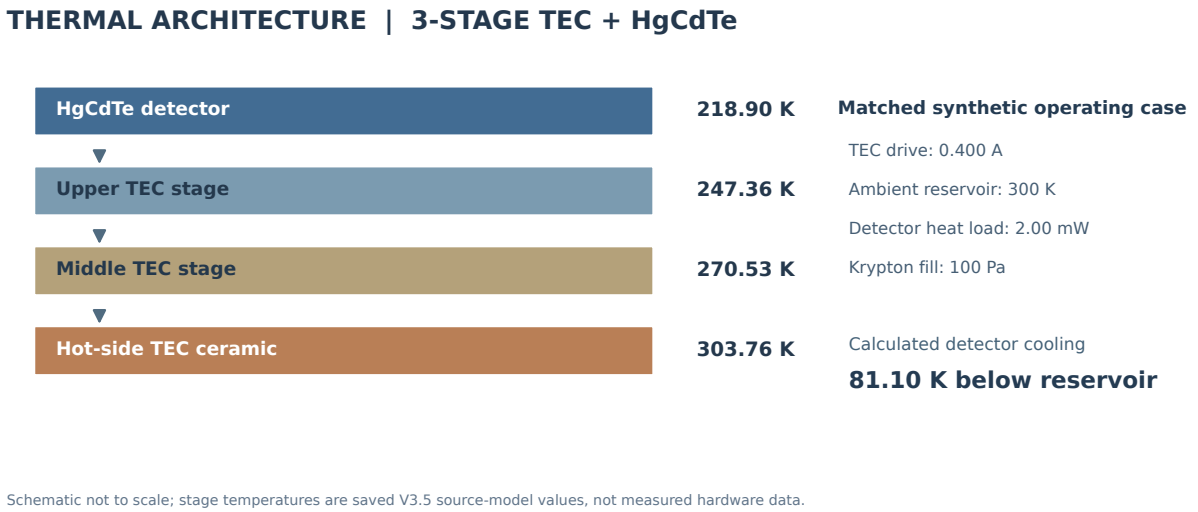
Thermoelectric Cooling of an HgCdTe Detector

Device Design, Heat Flow, and Electrical Performance

Three-stage cooler in a nine-pin package | 9 November 2026

PHYSICAL SCOPE. A three-stage cooler with a HgCdTe detector is evaluated using literature-informed thermoelectric properties, explicitly chosen package and joint assumptions, and calculated heat and current flow. The results describe a synthetic reference device, not a measured commercial product. Interstage foil heat leakage is now included through a conditional, source-CAD-based electrothermal multiport. True three-dimensional metal current spreading, deposited-film heat flow and measured joint impedances remain unresolved.



Conceptual thermal architecture of the current V3.5 operating point, with calculated hot-side ceramic and detector temperatures. This is a source-data schematic, not a geometrically scaled rendering or measured device. The true formed-foil CAD is shown later in the cooler-layout section.

Evidence standard. Calculated performance, material-property assumptions, source dimensions, and unresolved physical losses are separated throughout. A low residual demonstrates energy accounting, not measured detector performance.

Engineering summary

Question. How cold can a compact, three-stage thermoelectric cooler hold an HgCdTe photoconductor when heat must leave through a real package, and what thermal or electrical losses most threaten its performance?

A synthetic cooler is examined with four alumina substrates, 44 P/N thermoelectric couples divided among three stages (29 hot-side, 11 middle, 4 cold-side), soldered contacts, patterned metallization, supported formed copper-foil electrical transitions, and a HgCdTe die mounted at the cold end. The device occupies a proposed nine-pin metal package with a window. The interior is *physically designed*, but its material grades and joint specifications are not measurements of a commercial cooler.

Nominal operating condition: 0.400 A applied TEC current; 300 K ambient thermal reservoir; 2.00 mW imposed detector heat; 100 Pa krypton in the enclosure; literature-derived P/N transport curves and selected contact/heat-sink parameters. The calculated operating point is

$$\boxed{T_{\text{detector}} = 218.900\ \text{K}} \qquad \boxed{V_{\text{TEC}} = 1.489\ \text{V}} \qquad \boxed{P_{\text{electrical}} = 0.596\ \text{W}}.$$

This corresponds to an *approximately 81.10 K detector-to-reservoir temperature difference*, not a validated maximum TEC temperature drop or measured commercial specification. V3.3 without the formed copper bridge gave 210.625 K, while V3.4 with interstage copper and a combined power-lead approximation gave 218.833 K. The current V3.5 model resolves both TEC supply leads on their actual ceramic contact faces and produces 218.900 K, a total **8.275 K** above V3.3 under the declared reference hypotheses. The original V3.3 cavity-off/on comparison remains useful but is a separate earlier model-generation experiment.

Most important physical qualification: The newer electrical design removes the long unsupported copper-bar arrangement in favor of short formed foil links and a three-foot return, with defined ceramic landing pads. The current thermal calculation includes individually modeled P/N legs, 3D heat spreading through the four ceramics, solder/contact *electrical* losses, and effective package heat exchange. V3.5 retains the actual formed copper bridges and their three-foot return as an energy-conserving *reduced* electrical/thermal network, and corrects the two package-pin supply wires to original ceramic face 8 and 36. Its nominal Cu-lead temperature is 218.900 K. The nickel-like lead hypothesis differs by only +0.015 K, while a deliberately severe Cu interstage thermal-interface resistance changes the nominal result by **-6.276 K**. The extreme contact case is a sensitivity, not a recommended solder design. True 3D metal current crowding, solder thermopower and measured material/contact properties remain unqualified. Neither the former 210 K result nor the new 219 K result constitutes hardware validation.

Engineering use: The calculation identifies where energy is pumped, dissipated, parasitically imported or rejected, including about 110.6 mW removed from the hotter substrate through the V3.5 formed copper interstage connections, of which 87.2 mW enters the upper substrate; it predicts temperature, voltage and stage-by-stage heat loads under declared assumptions; and it provides a reusable starting point for evaluating metallurgy, foil geometry, contact thermal resistance, gas pressure and heat-sink design. Numerical mesh details and document-history evidence appear *after* the physical assessment.

Expanded TEC/TO design studies (9 November 2026). New separately audited source-architecture results examine matched P/N material transport screens, finite TO mounting resistance and matched-temperature TEC power, high-temperature transport-law continuation, and eight cavity gas/pressure and current cases. At 0.400 A and 30 kPa, the predicted top ceramic is 218.845 K with Kr, 219.277 K with Ar and 219.703 K with nitrogen; these values use a declared effective 3 mm gap, not a native three-dimensional cavity solution. The original geometric, electrical, temperature-field and numerical appendices are retained.

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1 Terminal wires, soldered TO pins, and thermal uncertainty

The TEC power supply must reach the miniature cooler through real wires soldered to the metal-package feedthrough pins. As supplied, these wires may be substantially longer than the *installed* length and are cut to fit the TO header. Only the final internal conductor from the cooler landing to each package pin contributes to the source ceramic-to-pin heat and current path; factory surplus outside the completed joint does not add thermal length to that connection.

The current source-derived layout has two independent, 0.200 mm diameter power leads. Pin 1 is connected to original ceramic face **8** over a 3.451 mm route; pin 9 is connected to original ceramic face **36** over a 4.965 mm route. The V3.4 average-hot-temperature model incorrectly combined both heat loads onto face 36. In the original saved ceramic field the two landings were near 303.94 and 298.39 K, respectively; they therefore need not exchange heat in the same direction with the 300 K header.

At *matched* 0.400 A, 300 K external reservoir, 2 mW detector heat and 100 Pa krypton, the improved three-stage calculations give:

Selected physical hypothesis	Detector (K)	TEC V	Input (mW)
V3.4 legacy lead map	218.833	1.48944	595.776
V3.5 Cu native pad	218.900	1.48917	595.668
V3.5 Ni-like wire	218.915	1.49525	598.100
V3.5 high Cu joint R	212.624	1.51877	607.509

The source-corrected copper-wire model predicts **218.900 K**, 0.067 K above the original V3.4 218.833 K. A same-geometry pure-nickel-like assumption raises that value by **0.015 K**. In contrast, raising the *interstage formed-copper joint* thermal-interface areal resistance from 10^{-6} to an extreme 10^{-4} m^2K/W lowers calculated die temperature to **212.624 K**. This is a deliberately severe interface hypothesis, not measured solder behavior or an endorsement of unreliable joints.

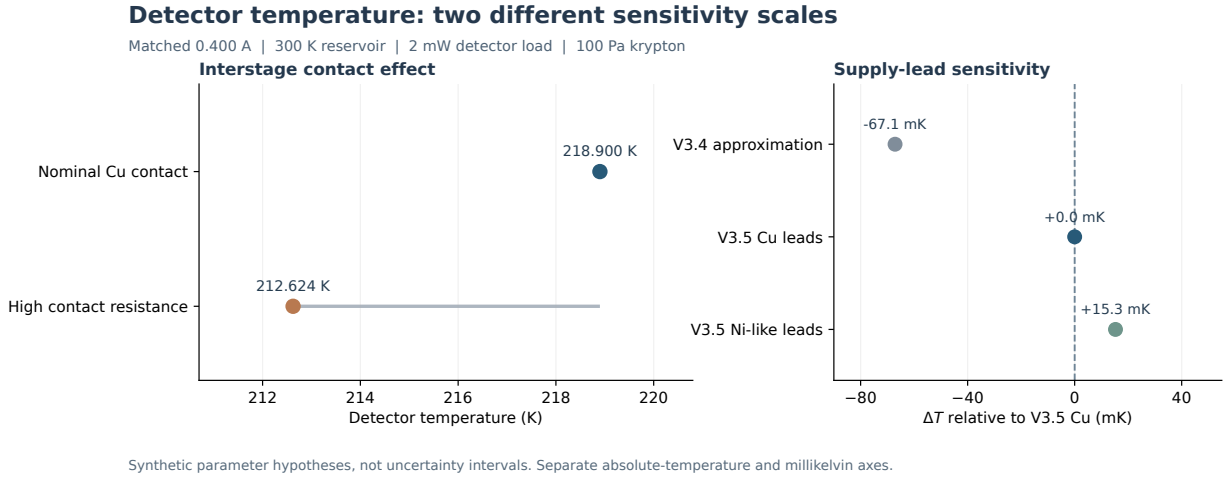


Figure 1: Two explicitly different horizontal scales for accepted full-system simulations at the same 0.400 A, 300 K, 2 mW and 100 Pa condition. Left: 6.276 K sensitivity to an extreme Cu interstage thermal-contact hypothesis. Right: millikelvin departures associated with the corrected terminal-lead model and selected Cu/Ni-like material properties. These are hypothetical design sensitivities, not measured uncertainty intervals.

1.1 Separate source contact heat, lead material, and cut length

For lead i with source internal length L_i, radius r_i, electrical resistivity ρ_i and thermal conductivity k_i, use

$$A_i = \pi r_i^2, \quad R_i = \rho_i L_i / A_i, \quad G_i = k_i A_i / L_i, \quad Q_{i\to c} = G_i(T_{\text{pin}} - T_{c,i}) + \tfrac{1}{2} I^2 R_i.$$

The one-half lead Joule partition is a reduced axial model assumption; the remaining half is rejected at the external pin. All terminal electrical I^2R enters the TEC voltage/power balance, without double counting the externally rejected half as cold-side heat.

Fully solved wire case	Pin 1 heat (mW)	Pin 9 heat (mW)	Wire Joule (mW)
Copper	-11.873	+1.948	0.737
Nickel-like	-2.446	+1.309	3.000

Signed heat is positive *into* the ceramic. With 300 K pin temperatures, pin 1 removes heat from its warmer ceramic while pin 9 imports heat to a cooler landing. Thus calculating both leads from one average hot-side temperature loses genuine local heat-flow information.

Nominal copper properties are $k = 400$ W/(m K), $\rho = 1.72 \times 10^{-8}$ Ωm; the illustrative nickel-like case uses $k = 90$ W/(m K), $\rho = 7.00 \times 10^{-8}$ Ωm. Actual manufacturer supplies should not be identified by color or visual impression: TEC Microsystems and RMT specify *tinned copper* as default, with manufacturer-dependent terminal options. The FeCrAl/Kanthal-like resistance-wire case is included only as a fixed-field counterfactual: at 0.400 A and the same 0.200 mm diameter geometry, its assumed resistance is 0.3884 Ω, generating *62.15 mW* of electrical heat. No such lead alloy has been identified on this physical cooler.

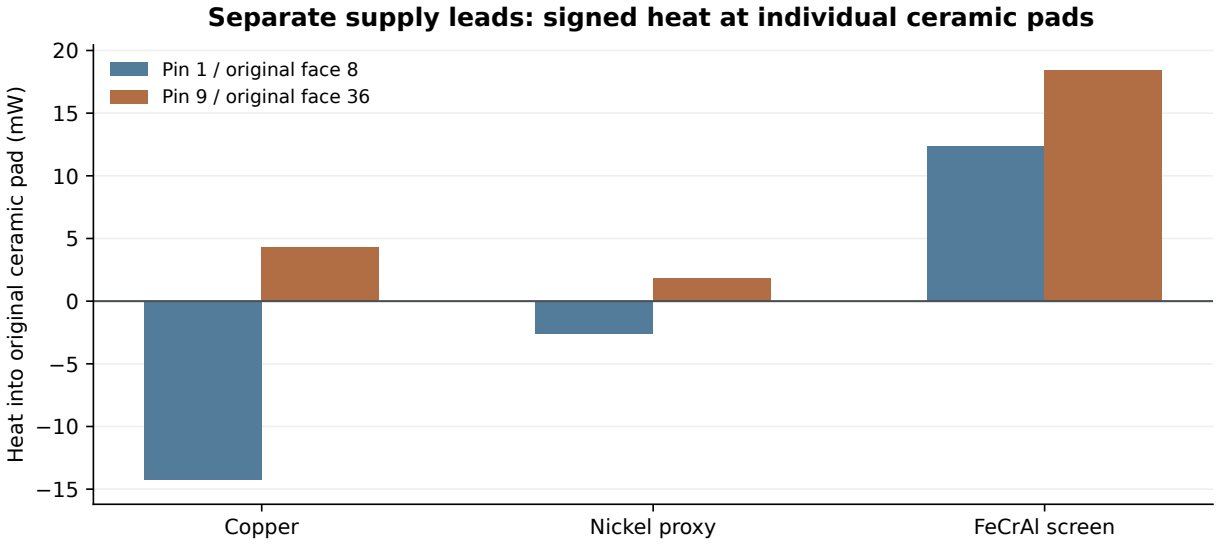


Figure 2: Lead material comparison at one frozen real V3.4 3D ceramic temperature field. Unlike the operating-point table above, this diagnostic is a fixed-field thermal/electrical source screen, not a nonlinear detector-temperature prediction.

1.2 The larger uncertainty: interstage copper and ceramic joint conductance

The three formed stage bridges carry both electrical current and parasitic heat. At each source-defined foil foot of area $A_c = 0.060$ mm^2, the provisional thermal contact conductance is $G_c = A_c / r''_{T,c}$. The nominal $r''_{T,c} = 10^{-6}$ m^{2}K/W is a selected non-measured contact assumption. A higher areal resistance reduces the conductive bypass in this model, but may be impractical or conflict with electrical continuity, solder fatigue, and actual fabrication.

Holding the *original* V3.4 source pad temperatures fixed, the first-pass upper-ceramic copper heat is:

Assumed $r''_{T,c}$ (m^2 K/W)	Upper-plate Cu heat (mW)
1e-07	90.611
1e-06	86.989
1e-05	62.664
1e-04	17.493

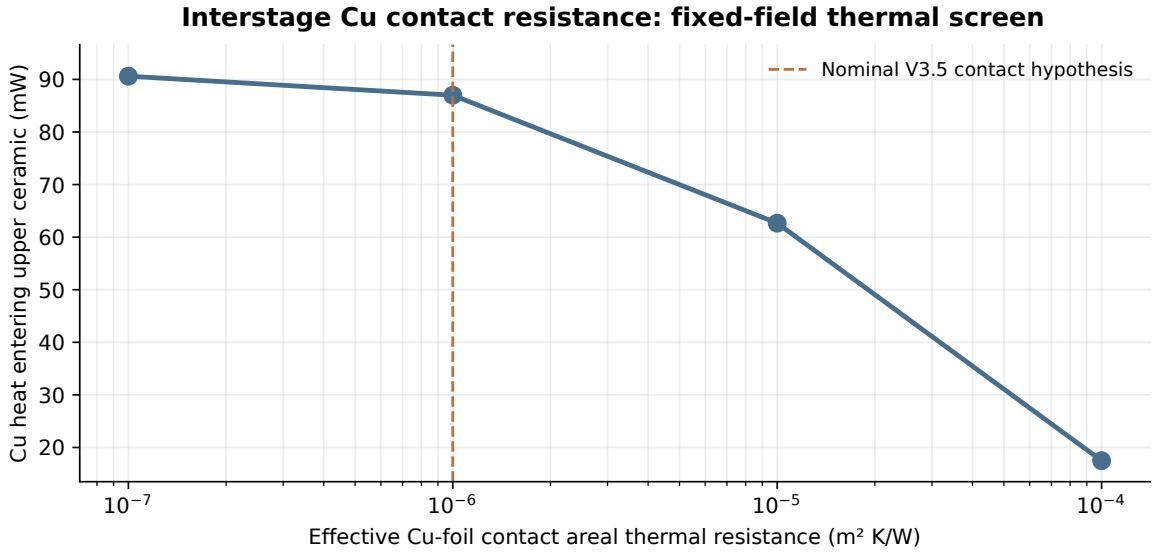


Figure 3: Source-pad-fixed heat-flux response to copper-foil interface assumptions. These intermediate screening fluxes are deliberately distinct from the three accepted fully coupled V3.5 operating-point results.

In the full-device copper-reference calculation, the upper plate receives **87.173 mW** through the Cu network and its hot stage-3 boundary is **247.359 K**. Under the severe interface hypothesis, those values become **21.697 mW** and **238.856 K**; the detector then equilibrates to 212.624 K. An 0.015 K lead-material change versus a 6.276 K copper-joint sensitivity identifies the next physically important question: *what is the realistic electrical and thermal interface law for the formed Cu foil, its solder feet, and the plated ceramic?*

1.3 Independent verification, limitations, and model use

Each V3.5 case uses the original 88 nonlinear axial P/N legs and four spatial ceramic FEMs with native source face heat. Three matched reference conditions converged with the following independent checks:

Scenario	Max pad update (K)	Stage residual (W)	Global residual (W)
Copper	1.767e-07	1.083e-09	9.376e-10
Nickel-like	2.441e-07	1.410e-09	1.205e-09
High R	1.456e-07	5.633e-11	4.577e-11

The stricter requirements remain below 3×10^{-7} K for the spatial update and 10^{-8} W for independently reconstructed stage and global energy balances. A separate audit re-evaluates both wire conductances/resistances from the unmodified source CAD, all three Cu bridge metal/contact networks (including the third thermal return

port), all stage energy budgets and terminal voltage. The documentation and plots consume only frozen accepted results; no new PDE solve is performed during report generation.

The wire and Cu laws are declared approximations, not measured low-temperature transport, OEM process tolerances or an as-manufactured TO feedthrough temperature solution. Neither the 0.200 mm diameter nor the installed route exactly identifies a vendor assembly. The next physical priority is credible solder/joint thermal resistance, plated-film heat spreading and conductor temperature-dependent properties. Cut-to-size factory terminal leads should be assigned the *installed* ceramic-to-pin length; an off-device surplus must never be treated as extra insulation that improves TEC temperature.

2 Current versus cooling performance: an operating envelope (V3.6)

The previous V3.5 model demonstrated a realistic source-connected 0.400 A operating point, but a hardware engineer needs to know how detector temperature and dissipated electrical power change with *TEC drive current*. V3.6 varies only the imposed current and detector heat load, using the *same* 88 source P/N transport legs, 11,791 ceramic tetrahedra, copper-foil interstage links, actual terminal feedthrough connections, krypton cavity and mounting sink assumptions. No material law or geometry was fitted to produce this curve.

At 300 K mounting reservoir, 2.00 mW detector heat and 100 Pa krypton, four independently conserved operating points give:

TEC current (A)	HgCdTe T (K)	Terminal V (V)	Input (W)
0.250	238.568	1.00644	0.25161
0.400	218.900	1.48917	0.59567
0.550	205.992	1.93574	1.06466
0.600	202.777	2.08091	1.24855

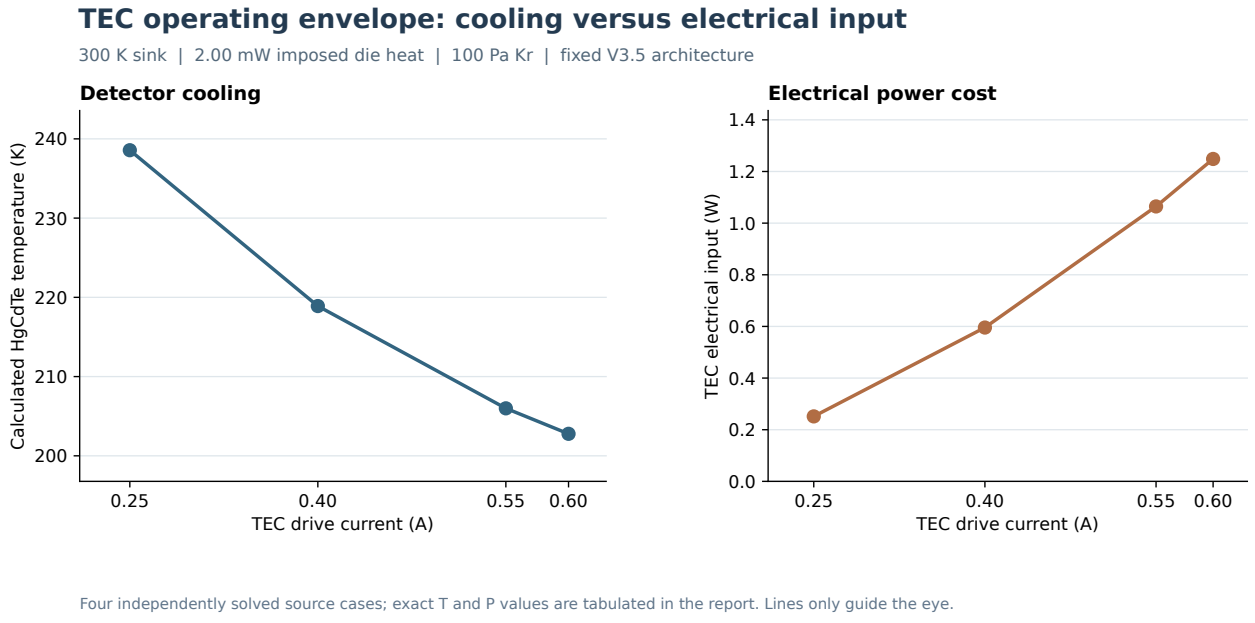


Figure 4: Four full-system source-model cases showing the decrease in detector temperature and increasing electrical input from 0.25 to 0.60 A. The line segments connect individually solved points; they are not a fitted continuous performance or safe-current curve. Other parameters are fixed at the nominal V3.5 physical hypotheses.

The upper-stage hot substrate cools from 260.43 K at 0.25 A to 237.36 K at 0.60 A, while the mounting-side TEC ceramic warms from 301.66 to 307.71 K because it must reject more heat through finite package impedance. The extra electrical cost of a further kelvin of detector cooling rises markedly:

Current step (A)	Cooling gained (K)	Extra input (mW)	Incremental mW/K
0.25–0.40	19.668	344.06	17.49
0.40–0.55	12.908	468.99	36.33
0.55–0.60	3.216	183.89	57.18

No minimum in detector temperature or safe current rating is claimed beyond the four tested conditions; increasing current still improved cooling at the upper tested 0.60 A boundary. The power budget is *module terminal input*, not the entire external power-supply wall consumption.

2.1 Sensitivity to real detector heating

At fixed 0.400 A, an imposed thermal load is an essential proxy for absorbed photons, bias heating and parasitic package heat before the HgCdTe transport physics is self-consistently integrated. The following points resolve the full TEC–ceramic–package feedback, not a single isolated thermal resistance:

Detector load (mW)	HgCdTe T (K)	TEC V (V)	Input (W)
1.00	218.589	1.48968	0.59587
2.00	218.900	1.48917	0.59567
6.00	220.145	1.48714	0.59485

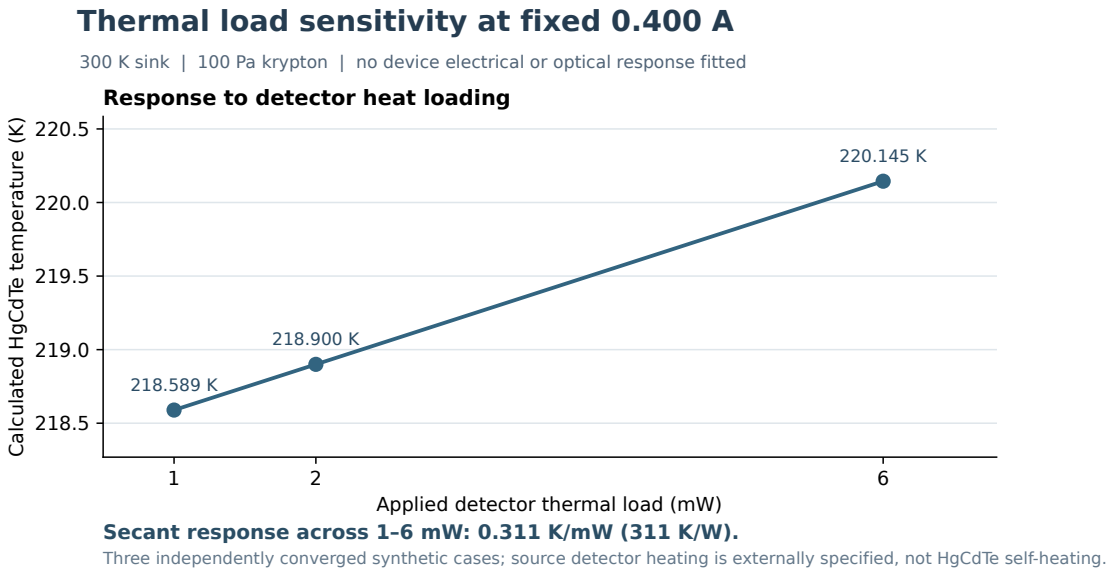


Figure 5: Source-based thermal load sweep at fixed 0.400 A, 300 K sink and 100 Pa krypton. The calculated full-device temperature changes by about 0.311 K per additional milliwatt across the tested 1–6 mW range. The HgCdTe resistance, signal and noise are not yet solved from semiconductor transport.

From 1 to 2 mW the observed secant is 310.734 K/W; from 2 to 6 mW it is 311.184 K/W. This approximately **311 K/W effective device thermal sensitivity** applies to the specified module and drive condition, not a measured intrinsic die thermal resistance. It demonstrates what the platform can now predict about tolerable bias/optical heat and electrical cost *before* an HgCdTe device-transport model supplies that heat self-consistently.

Numerical evidence and limitations. All six cases retain 88 nonlinear leg solutions, original four source ceramic fields and source-to-pin metal face heat accounting; separate no-solve audits recomputed each stage, copper, lead and global balance below 2×10^{-9} W. The unmodified 0.400 A/2.00 mW V3.5 reference is reproduced exactly. Two failed off-reference initializations were preserved and corrected by keeping the frozen 0.4 A heat budget during the first *warm start*; all physical equations and 3×10^{-7} K pad-update gates remain unchanged. These curves remain conditional on selected copper/solder, thermoelectric and mounting properties; no photoconductive gain, infrared detectivity or detector noise is inferred. All original earlier V3.3–V3.5 scientific results follow as independent historical evidence.

3 Thermoelectric transport sensitivity in the current cooler

An operating-temperature prediction depends critically on the actual paired P- and N-type material transport functions. The present synthetic cooler uses the full source-leg count, ceramic spatial fields, and metal heat paths, but its research-specimen $S(T)$, $\rho(T)$, and $k(T)$ curves are not a measured or manufacturer-qualified TEC material pair. It would therefore be premature to quote a narrow error bar on the absolute cold-stage temperature.

To quantify the leverage of these properties, a controlled source-geometry study changed one common P/N property factor at a time, with the original geometry, stage count, foil connections, mounting, gas, and electrical drive held fixed. Each point is a full 88-leg, four-ceramic nonlinear solution, *not* a scaled temperature inferred from an isolated thermocouple. The study applied $I_{\mathrm{TEC}} = 0.400$ A, a 300 K external reservoir, 100 Pa krypton, and a conditional auxiliary detector load obtained from $0.866025^2/R_{\mathrm{die}}(T)$ plus 0.500 mW absorbed heat. The illustrative $R_{\mathrm{die}}(T)$ law in this study is *not* used to infer actual detector performance; the gas study later in this assessment retains a strictly fixed 2.000 mW imposed heat load. Consequently the two experiment families must not be interpreted as identical detector-loading conditions.

P/N property hypothesis	Device mount (K)	Top ceramic (K)	TEC input (W)
All source factors 1.00	218.900	218.807	0.595668
Thermal conductivity ×0.80	207.812	207.716	0.616656
Thermal conductivity ×1.20	227.627	227.536	0.579393
Seebeck coefficient ×0.90	225.833	225.741	0.562663
Seebeck coefficient ×1.10	212.515	212.420	0.631764
Electrical resistivity ×0.80	216.563	216.470	0.523689
Electrical resistivity ×1.20	221.313	221.221	0.668943

Under the *selected, unequal* factor intervals, the total device-mount temperature endpoint spans are 19.815 K for k, 13.318 K for S, and 4.750 K for ρ, respectively. The ranking is an engineering screening result, **not** a statistical sensitivity ranking: the factor ranges are deliberately different, and independently modifying only one material property can be physically incompatible with a real thermoelectric grade.

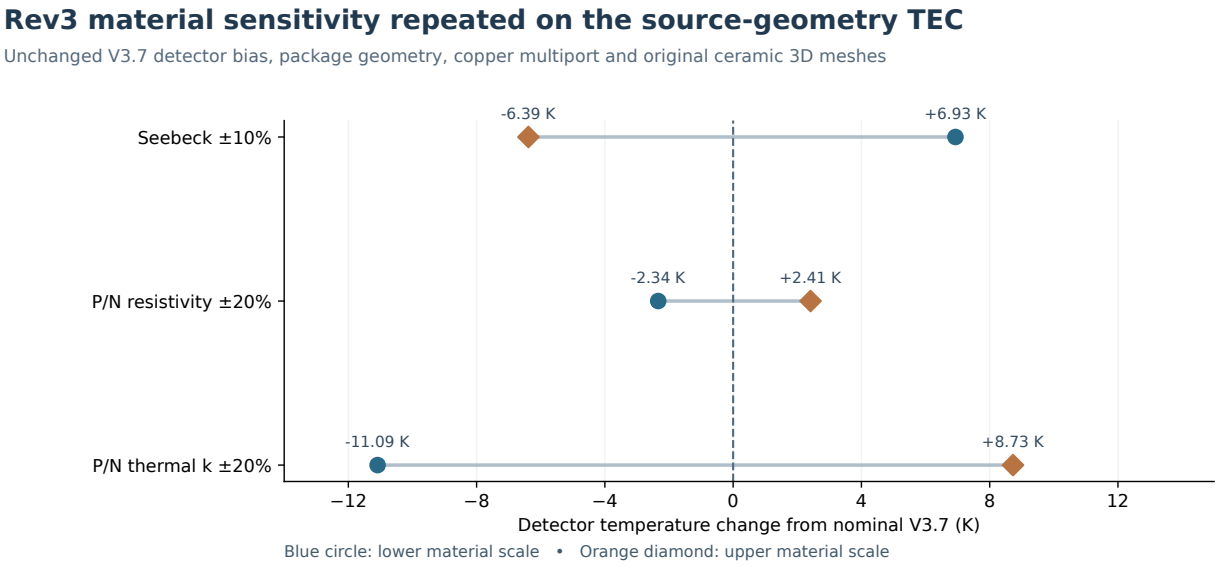


Figure 6. Independently solved P/N transport variations on the current full TEC/TO thermal architecture. Source-property factor ranges are hypotheses, not production tolerances.

4 Package mounting and power required to hold a cold temperature

Even with a fixed 300 K external reservoir, the hot ceramic is not constrained to 300 K. Heat is rejected through an assumed finite package mounting resistance. The nominal current reference uses $R_{\text{external}} = 4.833387$ K/W; the two controlled hypotheses add 5 or 10 K/W *on top of* that reference. The combined source-ceramic and 88-leg transport state was re-solved for each case, with the same stated reference detector thermal-input law and TEC current.

Additional R_{mount} (K/W)	Total R_{external} (K/W)	Device mount T (K)	TEC input (W)
0	4.833387	218.900	0.595668
5	9.833387	220.393	0.604935
10	14.833387	221.856	0.613656

The fixed-current temperature penalties are +1.493 K and +2.956 K for the added 5 and 10 K/W, respectively. Mounting resistance is a package design variable and should not be confused with the cooler's intrinsic material heat lift.

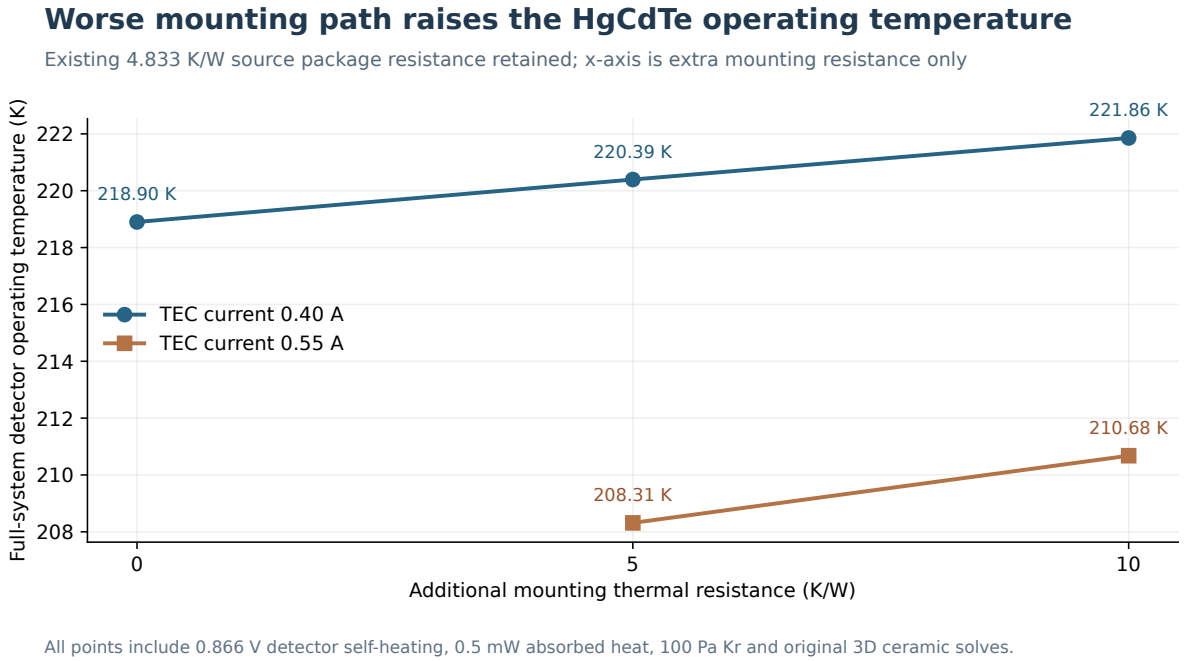


Figure 7. The finite external mounting path changes the nonlinear whole-device temperature distribution; the detector is not assigned a temperature rise algebraically.

A more practical procurement and instrument-design question is how much additional electrical power is required to *recover the same* operating temperature when mounting quality deteriorates. Two new registered, independently audited source solutions were obtained after a bounded current bracket and one numerical correction. These are *actual* full-system solutions, not interpolated electric power values.

Extra R (K/W)	TEC I (A)	Device mount (K)	Input (W)	Power penalty
0	0.400000	218.900024	0.595668	Reference
5	0.415286	218.892717	0.648375	+8.85%
10	0.433174	218.868845	0.712481	+19.61%

The maximum matched-temperature departure is just 0.0312 K. The +10 K/W mounting case demands approximately 116.8 mW more *TEC terminal power* to deliver essentially the same device-platform temperature, under the chosen boundary and material assumptions.

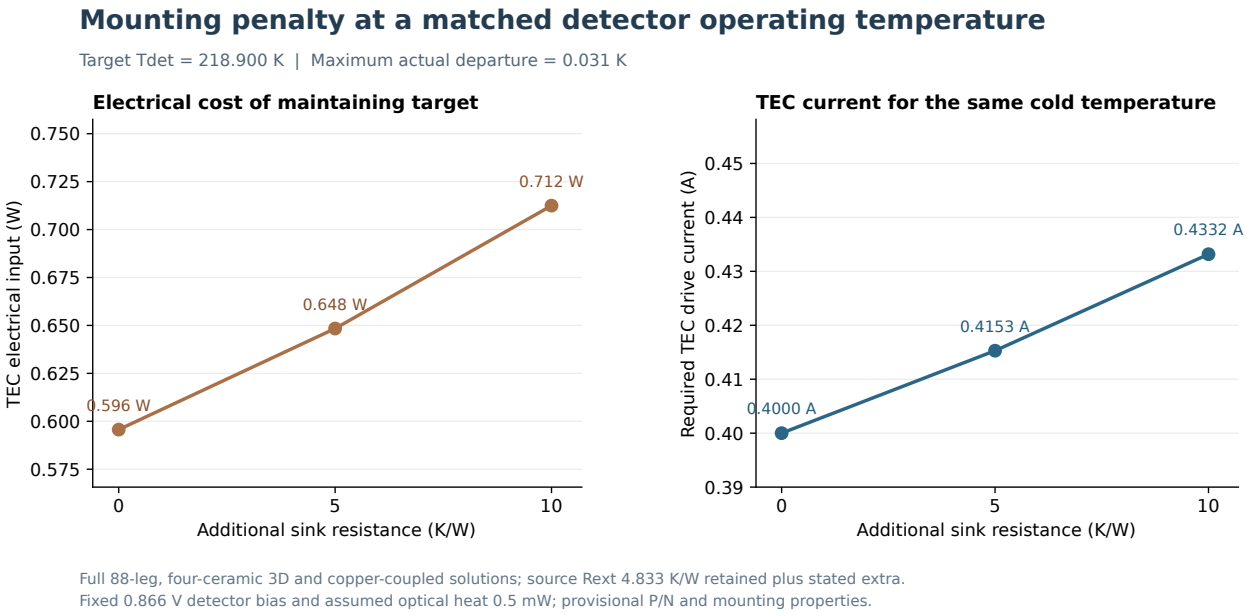


Figure 8. Calculated TEC current and terminal input required to maintain near-equal mounting temperature as external resistance increases. The comparison is not a measured OEM power rating or a global optimum.

4.1 Source-property validity above 300 K

At the reference mounting condition the hot ceramic reaches about 303.8 K; with an additional 10 K/W it reaches about 309.5 K. The P/N research curves have published markers through 300 K only. A separate six-run continuation check held the original 40–300 K curves exactly fixed, while assigning 0, 25, or 100% of the final source-curve tangent beyond 300 K. Across those hypotheses the device temperature spanned only 0.0170 K at nominal mounting and 0.0782 K under the worsened mounting condition. The corresponding total TEC input-power spans were 0.545 and 3.115 mW.

This result helps distinguish a *particular extrapolation parameter* from the much larger unqualified *bulk P/N material grade* question. It does not demonstrate that any of the three continuations is physically correct, nor does it provide a statistical uncertainty interval.

Engineering interpretation. Improving material-grade evidence and the mechanical/thermal TO mounting interface should precede an expensive broad numerical sweep of speculative high-temperature extrapolation functions. For any colder operating point, state current, terminal voltage, heat load, source-temperature coverage, mounting resistance, and the relevant electrical-power budget.

5 Effect of cavity backfill on the TEC top stage

The packaged cooler is also subjected to a potentially substantial heat leak from warm internal surfaces. A controlled eight-case study has now re-solved the *same* source-based TEC with rarefied-to-continuum Kr, Ar and N_2 backfill conditions. To separate cavity transport from detector-specific electronics, these simulations use a fixed $\boxed{Q_{\mathrm{die}} = 2.000\ \mathrm{mW}}$ rather than an electrical-resistance or optical absorption calculation. The principal comparison holds $I_{\mathrm{TEC}} = 0.400$ A and external temperature 300 K.

Backfill	Top ceramic (K)	Device mount (K)	Gas heat (mW)	TEC power (W)
Vacuum, gas off	218.353	218.437	0.000	0.596111
Kr, 100 Pa	218.807	218.900	3.420	0.595668
Kr, 30 kPa	218.845	218.939	3.724	0.595630
Kr, 101.325 kPa	218.845	218.939	3.725	0.595630
Ar, 30 kPa	219.277	219.380	6.966	0.595209
N_2, 30 kPa	219.703	219.815	10.170	0.594792
N_2, 101.325 kPa	219.703	219.815	10.172	0.594792

At the same 30 kPa nominal pressure, nitrogen is 0.8753 K warmer than krypton at the device mounting node and imports an additional 6.446 mW of gas heat. Argon lies between the two. Disabling gas conduction entirely gives a 0.4634 K cooler mounting condition than 100 Pa Kr, while leaving solid and radiative conduction active. No numerical result is assigned to a manufacturer or a known production-gas recipe.

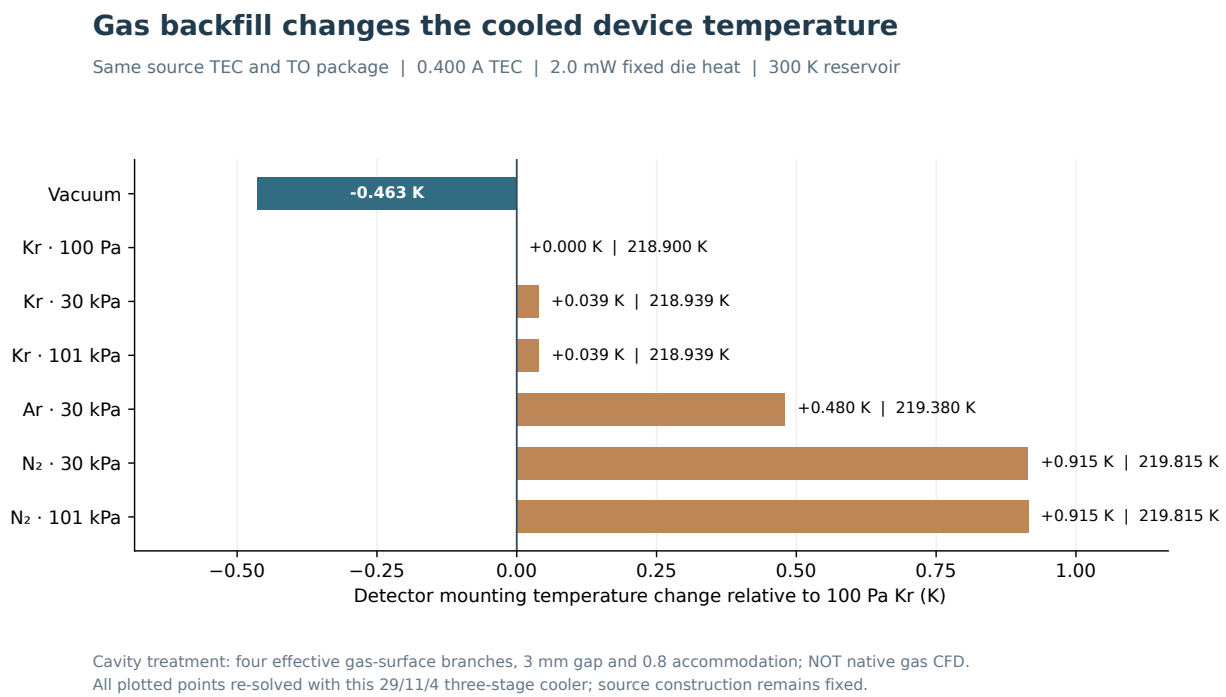


Figure 9. Backfill gas and pressure sensitivity of the device-side mounting temperature. The top TEC ceramic is a distinct lower-temperature node; both are listed in the accompanying table.

6 Gas heat-transfer mechanism and its uncertainty

The current cavity calculation is a *reduced effective-gap* conduction model, not a finite-volume or finite-element solution of gas transport around all 88 exposed thermoelectric legs. For a warm enclosure surface and cold branch i,

$$Q_{\mathrm{gas},i} = G_{\mathrm{gas},i}(T_{\mathrm{env}} - T_i), \qquad G_{\mathrm{gas},i} = \frac{k_{\mathrm{gas}}(\overline{T})A_i}{g + 2\ell_{\mathrm{jump}}(p, \overline{T}, \alpha)}.$$

Here $g = 3$ mm is an *assumed* effective gap, the thermal accommodation coefficient is $\alpha = 0.8$, and the four assigned branch areas are $12.60,\ 8.68,\ 4.76,\ 1.00$ mm^2. Temperature-dependent tabulated Kr, Ar and N_2 conductivities, an Eucken-based viscosity estimate, and a temperature-jump rarefaction correction determine each conductance.

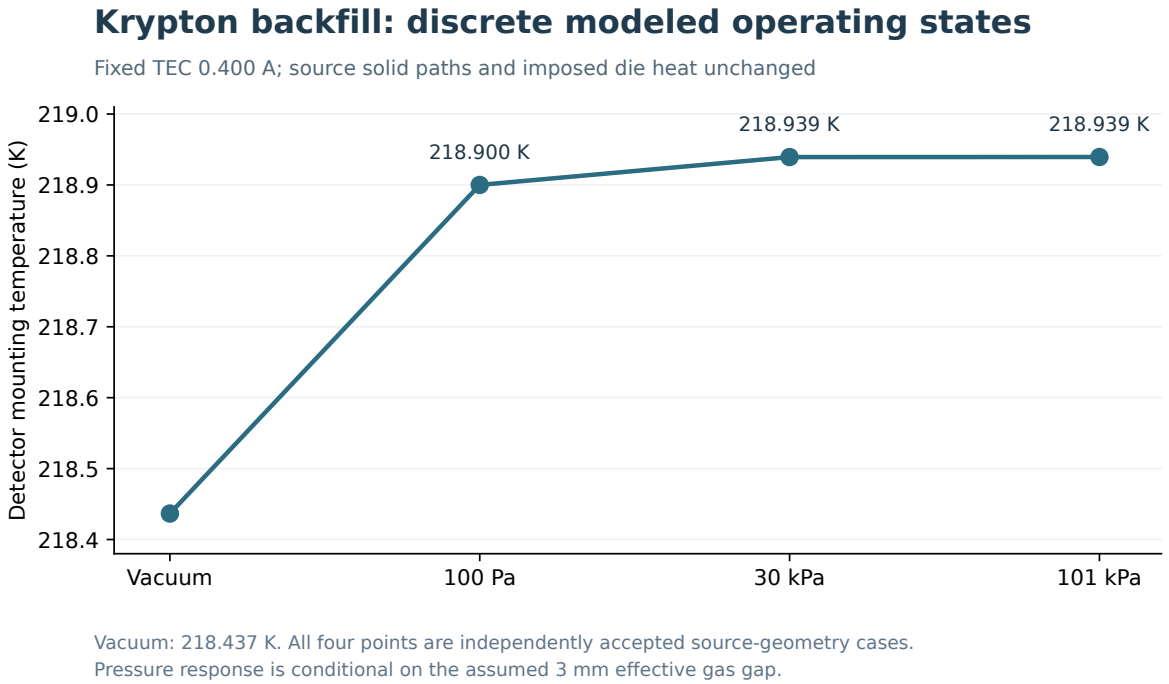


Figure 10. Discrete rarefied-gas/pressure scenarios for Kr in the declared effective-gap model. Connecting lines are guides only, not a measured or independently verified continuum response curve.

Under this particular model, raising Kr pressure from 100 Pa to 30 kPa adds just 0.03930 K to the device temperature; raising it from 30 kPa to 101.325 kPa adds only 0.00010 K. This apparent saturation is largely a consequence of the chosen effective gap, warm-area mapping, and accommodation law. The physically available narrow gaps between 0.40-mm-wide pellets spaced on a 0.58-mm pitch are not represented as independently exposed gas channels. A more faithful cavity model could change both the pressure transition and the gas species ranking. The present data **must not** be used to claim pressure is unimportant in an actual TO package at 100 Pa.

7 Thermal loading, electrical demand, and engineering optimization

Changing TEC drive current is not a gas effect, but it sets the trade-off between cold temperature and electrical power. A separately solved case at $I_{\mathrm{TEC}} = 0.550$ A and 30 kPa Kr, with the same 2 mW die heat and 300 K reservoir, gives

$$T_{\text{top ceramic}} = 205.931 \text{ K}, \quad T_{\mathrm{die}} = 206.032 \text{ K}, \quad P_{\mathrm{TEC}} = 1.06462 \text{ W}.$$

Relative to 0.400 A at 30 kPa Kr this lowers device temperature by 12.908 K while raising terminal input by 0.469 W. Such a trade-off is useful only when a real operating envelope and safe current/voltage limits are independently established.

7.1 Prioritize physical knobs before creating a broad optimization surface

The existing new-architecture studies support the following *engineering experiments*, with their distinction between real geometric variables and screening assumptions preserved.

Construction or operating knob	Existing conditional effect	Next physical qualification
P/N material transport k, S, ρ	19.8, 13.3, 4.75 K source-factor spans	Real matched P/N grade and correlated properties
External mounting resistance	+2.96 K at +10 K/W, fixed 0.4 A	TO contact patch, heat spreader and interface pressure
Formed-Cu foil/joint thermal path	6.276 K for an *extreme* contact hypothesis	Solder/plating area, Cu bending, electrical resistance and current crowding
Gas species, 30 kPa, Kr to N_2	+0.875 K	Native cavity exposure and inter-pellet gap distribution
Detector prescribed load, 1–6 mW	About 1.56 K span at 0.4 A	Define die/power budget, attach and carrier conductivity
TEC drive current, 0.4 to 0.55 A at 30 kPa Kr	−12.908 K at +0.469 W	Rated current, lifetime and allowable package dissipation

A physically credible optimization of top-stage temperature should vary the pellet area A, height L, stage placement, current and source-grounded package thermal boundaries *jointly*, subject to specified electrical and mechanical constraints:

$$\min T_{\mathrm{top}}(I, A, L, Q_{\mathrm{die}}, T_{\mathrm{sink}}, R_{\mathrm{mount}}, p_{\mathrm{gas}}, \ldots).$$

A transport-only A, L sweep can isolate competing $R_{\mathrm{leg}} \sim \rho L/A$ and $K_{\mathrm{leg}} \sim kA/L$ effects. It cannot by itself be called a fabricated pellet-geometry variant: changing width or height also changes solder pad area, foil contact location, ceramic spacing, interleg gas gaps and insulation clearance.

7.2 Evidence quality and outstanding physical work

All eight gas cases and the material/mounting studies archive original 88-leg nonlinear states, source native four-ceramic 3D nodal fields, input registrations and independent audits. The eight gas cases all pass direct independently reconstructed gas conductance, source Cu/lead Joule, three-stage and global heat balances; their maximum global energy residual is below 2×10^{-9} W. This demonstrates conservation *of the declared model*. It does not identify actual P/N pellet batches, solder thermal impedances, or exposed package gas paths, nor does it establish a full multi-material 430-domain current and thermal PDE.

The next high-value releases should therefore be: a source-grounded pellet width/height feasibility screen, a native exposed-surface cavity network, and a quantified TO mounting/contact construction study. For each, report *top ceramic temperature*, die mounting temperature, thermoelectric input, package rejected heat, and individual component heat-flow budgets. Semiconductor signal/noise and photodetection belong to the separate specialist model; they are not necessary to answer these package thermal-design questions.

8 Original V3.4 copper-bridge result and its physical explanation

A thermoelectric cooler transfers heat toward its hot side while electrical interconnects must also cross between ceramic stages. The three short formed copper-foil connections in the current cooler provide a real electrical path, but they simultaneously create passive conductive paths that partially undo the cooling. This effect is absent from the original V3.3 prediction. A new V3.4 calculation adds the actual two-foot and three-foot link topology with physically separate ceramic-pad thermal terminals.

At the *same* 0.400 A input, 300 K heat reservoir, 2.00 mW detector heat and 100 Pa krypton setting:

Calculated quantity	V3.3	V3.4 with copper	Change
Detector temperature (K)	210.625214	218.832895	+8.207681
Terminal voltage (V)	1.522462	1.489440	-0.033023
Electrical input (W)	0.608985	0.595776	-0.013209
Sink heat rejection (W)	0.594044	0.579830	-0.014214

Although the predicted electrical power falls, the detector is **8.208 K warmer**. The important loss mechanism is *parasitic stage-to-stage conduction* through the copper, not simply additional resistive power dissipated in the tabs.

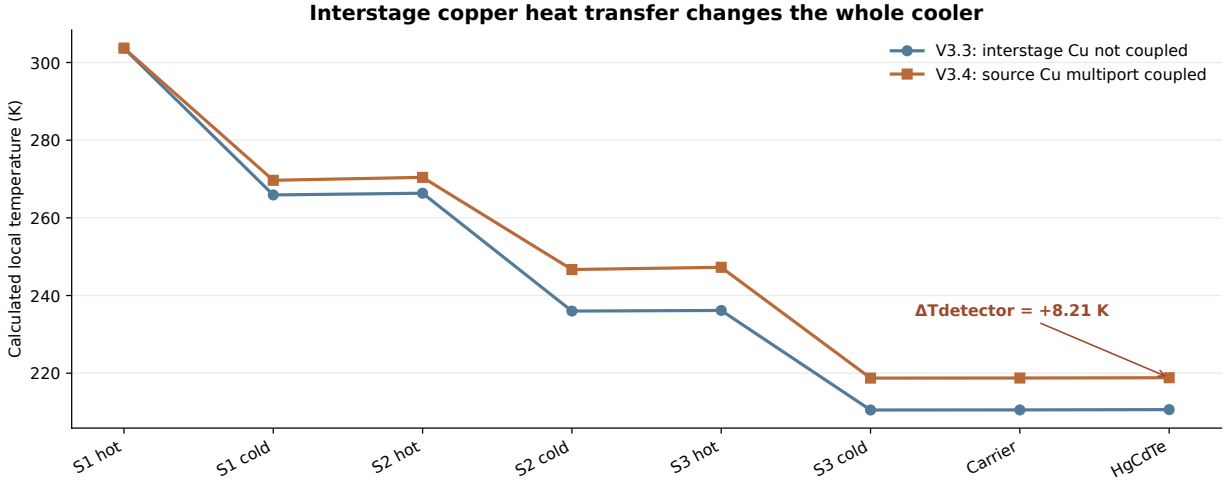


Figure 11: Solved source-ceramic stage and detector temperatures in V3.3 versus V3.4 at matched source assumptions. V3.4 adds a geometry-derived, finite-contact copper heat network; this is a conditional theoretical comparison, not measured or full 3D copper-metal fields.

8.1 Copper heat paths and their thermal consequences

The original CAD includes copper-foil tabs across electrical nets 58 and 80 and a stepped return at net 88. Their original contact sites on the three source ceramic planes are:

$$\text{net58}: \ 38 \to 75, \qquad \text{net80}: \ 79 \to 105, \qquad \text{net88}: \ 36 \to 74 \to 104.$$

The actual bridge foil width/thickness are 0.300/0.030 mm. Heat transfer uses the actual *center-to-center* solder-foot spans, not the full unsupported length of each tab. Net88 has an intermediate supported landing and therefore three separate thermal terminals. The model does not treat that landing as insulated.

For a foil segment i, j and finite contact c, i:

$$G_{ij} = k_{\mathrm{Cu}} A / L_{ij}, \qquad R_{ij} = \rho_{\mathrm{Cu}}(T_{ij}) L_{ij} / A, \qquad G_{c,i} = A_{c,i} / r''_{T,c},$$

where the provisional material laws use $k_{\mathrm{Cu}} = 385\ \mathrm{W/(m\,K)}$, $\rho_{\mathrm{Cu}}(T) = 1.724 \times 10^{-8}[1 + 0.0039(T - 293.15)]\ \Omega\,\mathrm{m}$, $r''_{T,c} = 10^{-6}\ \mathrm{m^2K/W}$, and $r''_{e,c} = 10^{-11}\ \Omega\,\mathrm{m^2}$. These are **selected engineering hypotheses**, not identified foil or solder-lot properties.

The node equation

$$\sum_j G_{ij}(T_i^{\mathrm{foil}} - T_j^{\mathrm{foil}}) + G_{c,i}(T_i^{\mathrm{foil}} - T_i^{\mathrm{ceramic}}) = P_{J,i}$$

sends local conduction and Joule heat to the original 3D ceramic-pad faces. The signed results below are reported as *heat into ceramic*; negative values indicate the foil removes heat:

Foil connection	Hot (mW)	Mid (mW)	Upper (mW)	Joule (mW)
Net 58	-54.594	+55.141	+0.000	0.547
Net 80	+0.000	-40.619	+41.115	0.496
Net 88	-53.881	+9.044	+45.875	1.038
ALL	-108.475	+23.566	+86.989	2.080

The copper network removes approximately 108.5 mW from the hot ceramic while depositing 87.0 mW in the upper ceramic. These are *redistributed* heat flows, not net external heat creation. Their stage total equals the copper circuit's 2.080 mW of electrical Joule dissipation.

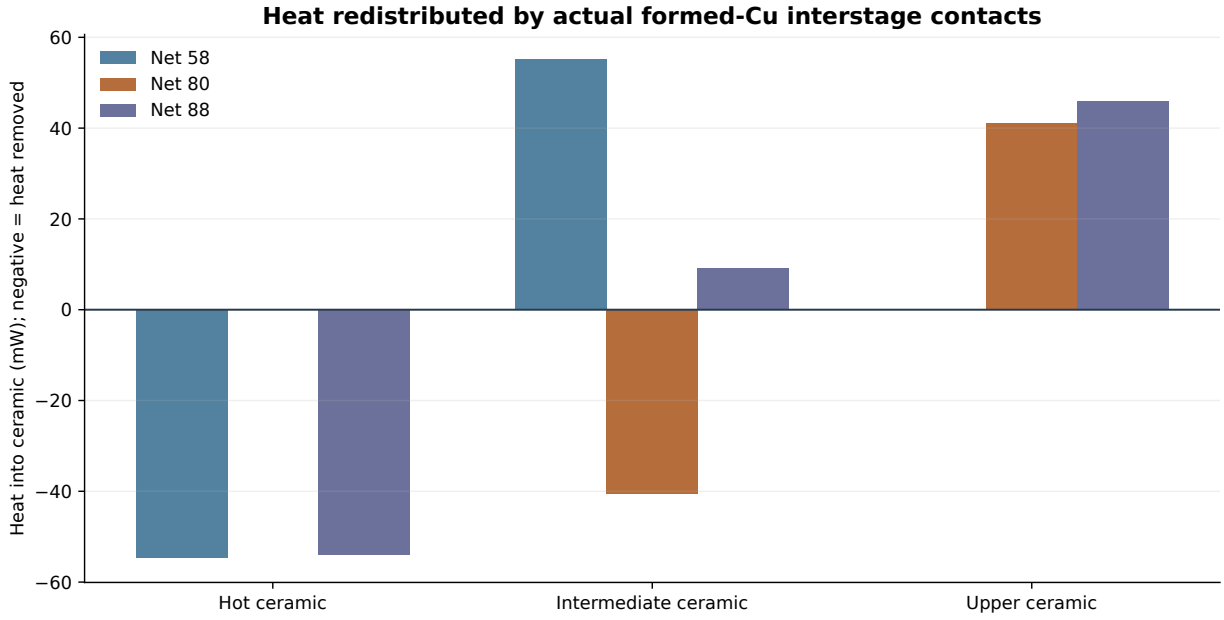


Figure 12: Original-CAD-face copper heat flows from accepted V3.4: individual networks at nets 58, 80 and 88. Positive denotes heat entering ceramic. Net88's middle contact has its own nonzero heat flux.

8.2 Resulting three-stage heat pumping and numerical checks

TEC stage	Number of legs	Q_c (mW)	Q_h (mW)	Voltage (V)
1	58	287.265	709.342	1.055194
2	22	136.178	260.411	0.310583
3	8	4.910	46.991	0.105201

The upper-stage hot boundary increases from 236.16 K (V3.3) to 247.27 K (V3.4). The stage therefore operates across a materially altered thermal environment while still providing approximately 4.910 mW of cold-side heat pumping.

Two earlier fixed-point attempts remain archived as **STOP**; the second passed a coarse pad-temperature threshold but failed the final energy test. The accepted third attempt meets the tighter 3×10^{-7} K temperature-update criterion after **19** outer steps, without changing the source physics:

Independent engineering check	Absolute residual
Final pad update (K)	1.399e-07
Independent stage heat residual (W)	7.959e-10
Global energy residual (W)	6.775e-10
Copper Joule/heat mismatch (W)	8.405e-16

A separately implemented read-only audit reconstructs copper Joule, all three stage heat balances and the hot-reservoir balance from the saved original-face and node data, without rerunning the coupled PDE. The 3D FEM still solves *only the ceramics*; the copper model remains a source-connected 1D heat/electric network with assumed contacts. These residuals establish internal conservation, not an experimental uncertainty interval. The principal next uncertainty is whether actual copper conductivity, foil/joint resistance, plated sheet spreading and geometry tolerances preserve the calculated 8.2 K shift.

9 Device construction and thermal architecture

Four nominal ceramic plates have square footprints 6.4, 5.2, 3.8 and 2.4 mm, each 0.5 mm thick. Eighty-eight P/N legs are 0.4 by 0.4 by 0.5 mm each, grouped into 58, 22 and 8 legs from hot to cold. The TEC supply is connected to header pins 1 and 9; the HgCdTe photoconductor has separate bias contacts on pins 7 and 8. The gold-colored contact patterns on the ceramic are internal series metallization and solder lands, *not* the external positive and negative power terminals. The exported reference contains 824 CAD solids and 176 solder joints, 89 insulated TEC electrical nets, and formed internal metal links. The original long, unsupported copper routing was rejected. The replacement has formed Cu-family foil tabs across nets 58 and 80, plus an extended three-foot return net 88 with a real middle thermal landing. Its design uses patterned ceramic metallization and finite solder footprints. The selected foil section is 0.300 by 0.030 mm; its bend radius is 0.075 mm; finished foot joints are specified at 20 plus/minus 5 micrometres. No measured current rating or fatigue qualification follows.

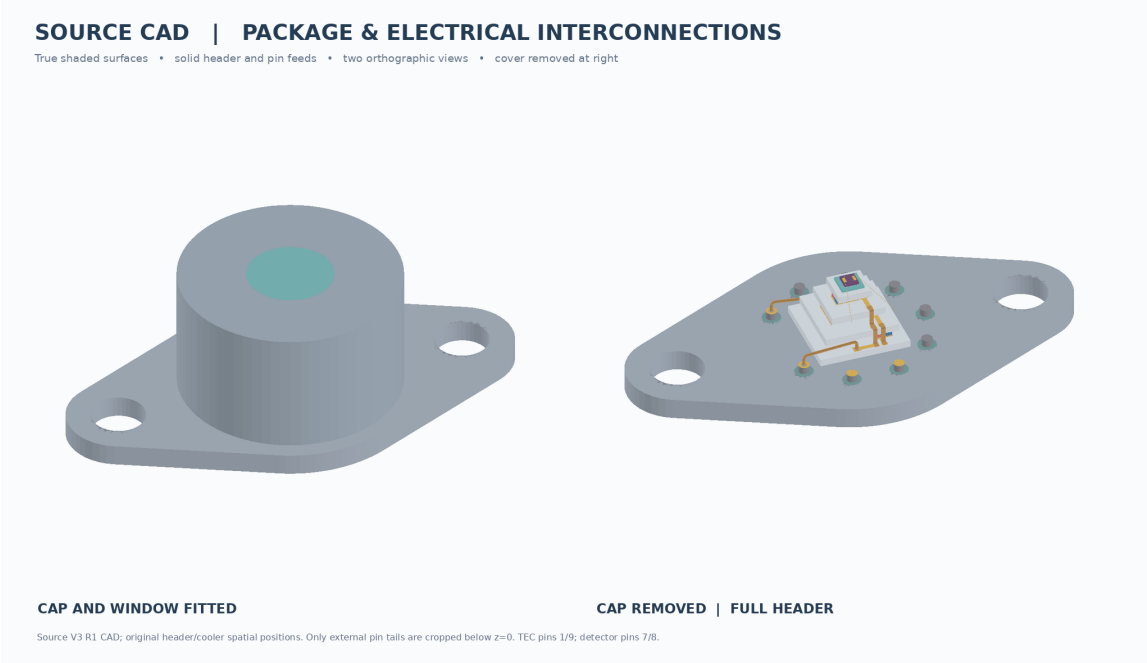


Figure 13: Physical arrangement of the proposed packaged HgCdTe device. Left: closed metal cap and optical window above the nine-pin header. Right: identical source-derived assembly with cover removed, exposing the thermoelectric stack, foil connections, TEC supply leads, and separate detector wiring. The external lead tails below the header are cropped for clarity; the mechanical assembly is synthetic, not a manufactured module.

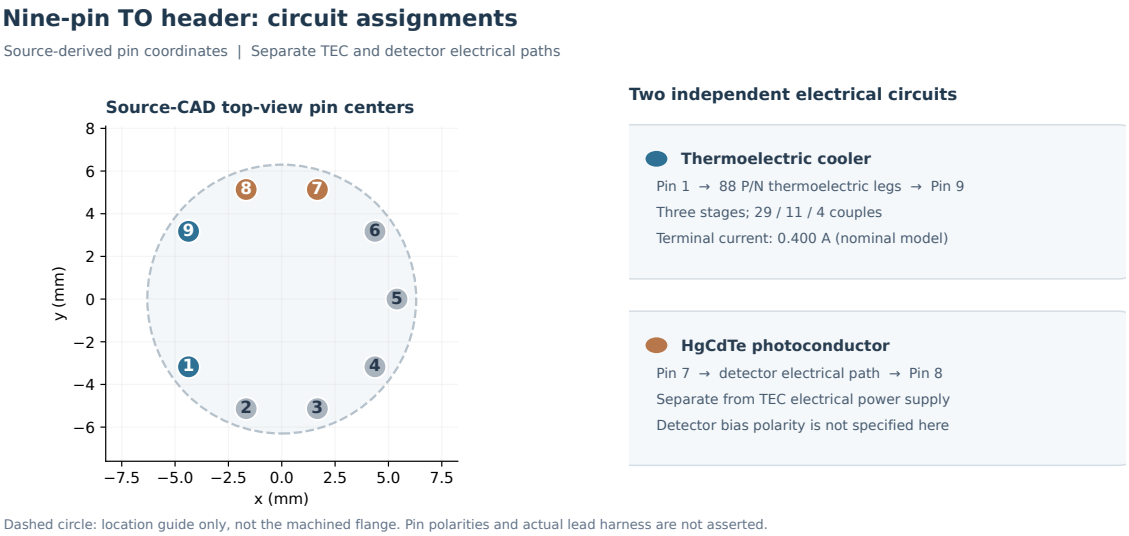


Figure 14: Source-registered physical top-projection of all nine header feedthrough centers in millimetres. Blue pins 1 and 9 connect the thermoelectric series path; orange pins 7 and 8 connect the separate HgCdTe photoconductor circuit. The pin-location guide is not the real machined header outline. No unverified polarity or wire-harness assignment is implied.

10 Cooler layout and supported interstage connections

The shaded views below tessellate the original 824-body source geometry using a true per-pixel depth buffer. The previous mean-face-depth renderer wrongly displayed hidden metallization through ceramic solids. This rendering is neither a reconstructed cooler nor an exploded assembly. All plates, P/N legs and analytically bent copper foil links retain the source positions, widths, heights, polarities and original path radii. Compared with the wireframe, the opaque solid bodies make the differences from the rejected long copper-post construction readily visible.

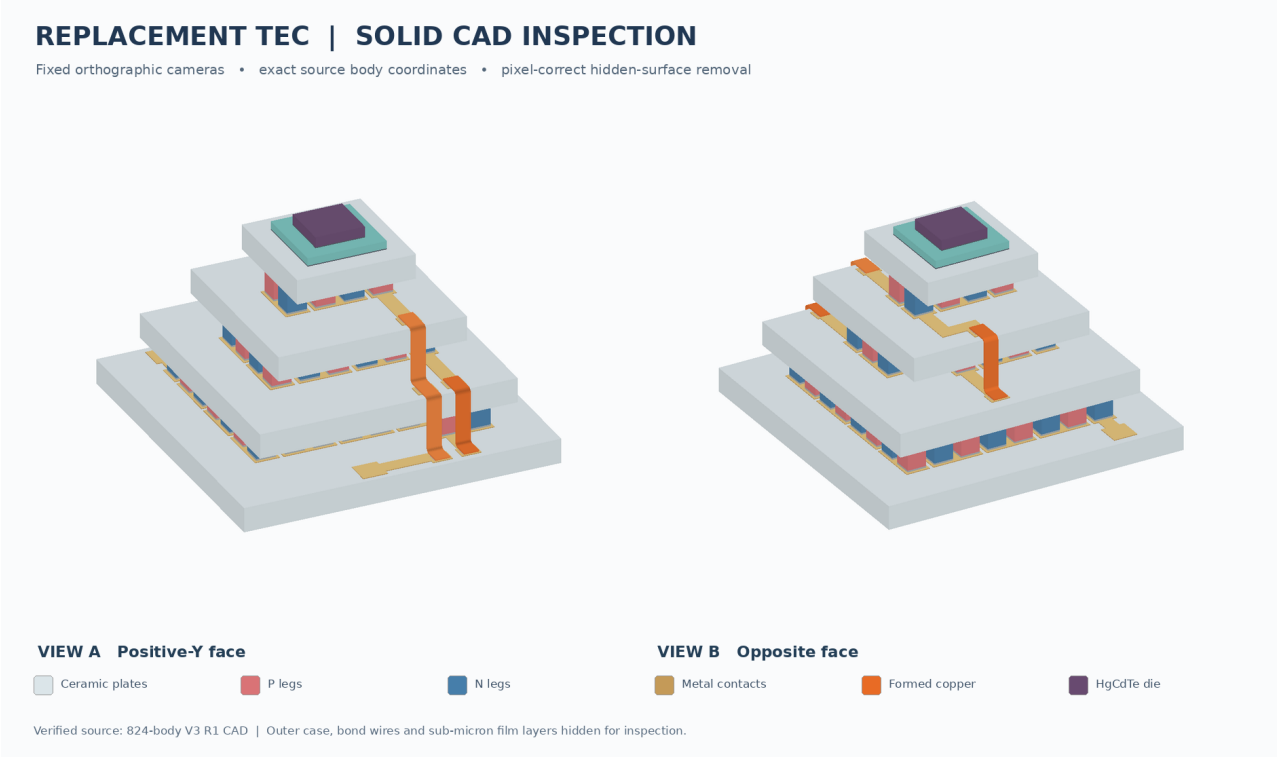


Figure 15: Correctly depth-occluded replacement geometry from two orthographic views: four ceramic stages, 58/22/8 P/N legs, visible exposed metallization only, and actual formed copper-foil connections. The external case, leads and very thin buried film layers are intentionally omitted. Original analytic foil-bend paths are sampled for shaded display, without relocating any CAD features.

10.1 Actual interstage foil cross-sections

The design uses two short supported formed copper-foil stage links (nets 58 and 80), and a longer stepped return (net 88). The last has two vertical steps and three physical feet, including an intermediate thermal landing. Unlike a generic connecting rod, its cross-section has nominal width 0.300 mm, thickness 0.030 mm and bend radius 0.075 mm. The foil is represented at actual thickness, not enlarged for visibility.

This original-CAD visualization establishes the mechanical design, rather than a simulated copper current or temperature field. V3.4 and V3.5 now account for foil conduction, Joule loss and finite metal/ceramic thermal contacts with a conservative reduced multiport. True three-dimensional current crowding, metallization-sheet spreading, measured joints and actual fabrication remain outside the qualified model.

11 Electrical routing and interstage heat leakage

The CAD conductor graph verifies all 88 legs as articulation vertices in the pin-1-to-pin-9 series circuit: removing any one leg disconnects the two terminals. Native transitions pass from P leg 57 to N leg 58 through tab 58, and from P79 to N80 through tab 80. The final leg P87 is connected to pin 9 through stepped return 88. These checks prove intended topology but not the temperature-dependent electrical or thermal properties of the connections. For a uniform slender conductor with temperature-dependent material properties,

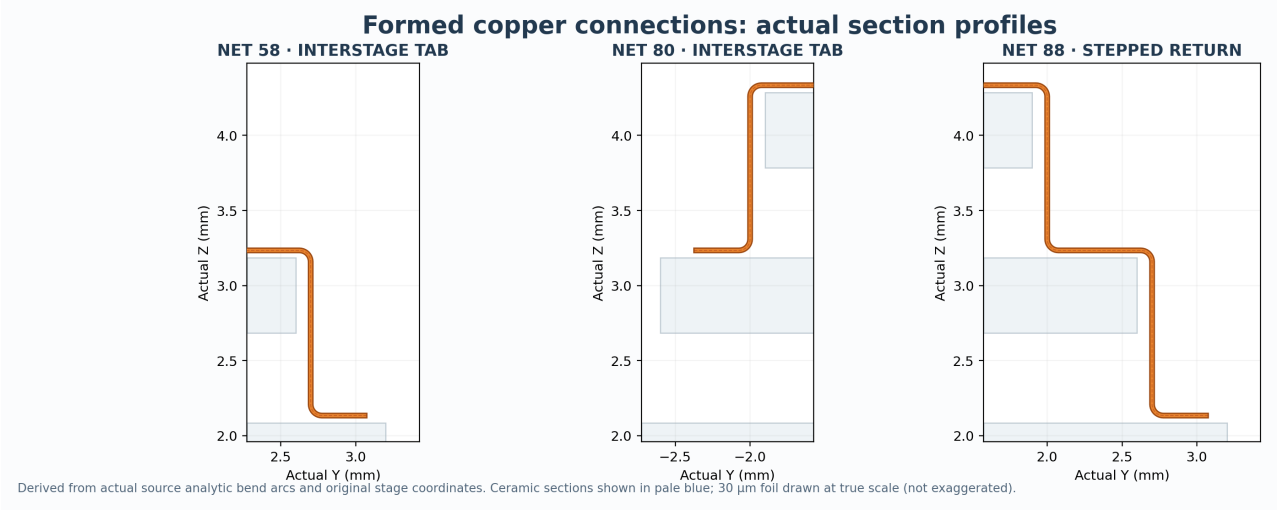


Figure 16: Y–Z sections reconstructed from the source analytic straight and circular-arc geometry at original coordinates, without exploded offsets. Pale blue regions indicate where ceramic cross-sections intersect each foil plane. The 30-micrometre copper thickness is drawn at true scale.

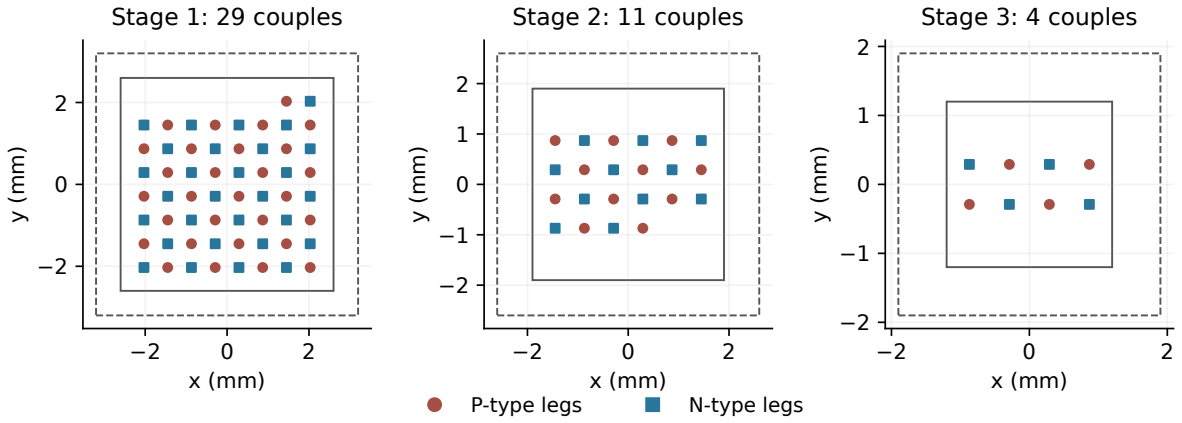


Figure 17: Actual saved stage-specific P/N leg centroids and plate footprints. This verifies the source positions used in the working hybrid; it does not depict a solved copper-tab thermal field.

$$R_e = \int_0^L \frac{\rho(T,s)}{A(s)}\,ds, \qquad G_{\mathrm{th}} = \left[\int_0^L \frac{ds}{k(T,s)A(s)}\right]^{-1}, \quad P_J = I^2 R_e.$$

The registered neutral path lengths are 1.96062 mm (net58), 1.97062 mm (net80), and 3.70624 mm (return88). Using a uniform 0.300 by 0.030 mm section, illustrative $\rho_{\mathrm{Cu}} = 1/(58\times10^6)\,\Omega\mathrm{m}$ and $k_{\mathrm{Cu}} = 385\,\mathrm{W/(m\,K)}$ yields the approximate geometry-only screen below. The competing design goals are explicit: reducing electrical resistance

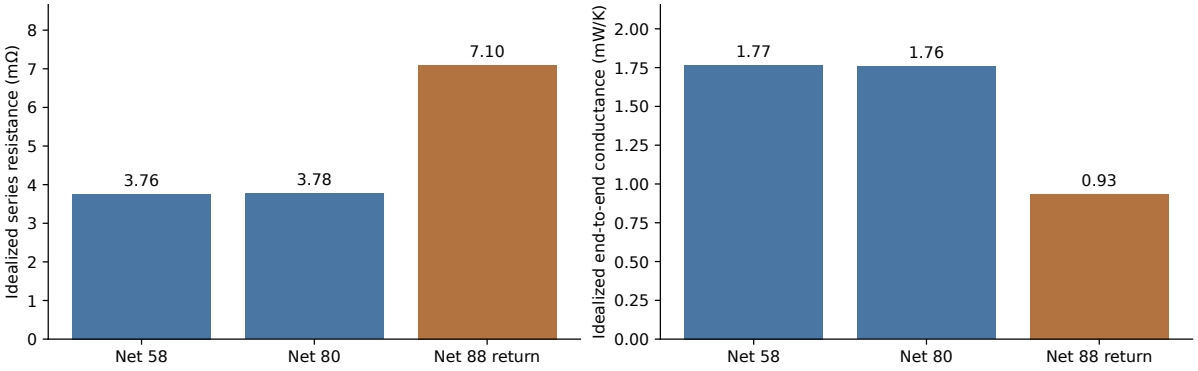


Figure 18: Historical pre-V3.4 screening of idealized uniform-bar electrical and thermal copper coefficients. Actual formed-tab bends, patterned multilayer film, contact drops, variable temperatures, Joule partition and intermediate return landing are not resolved. The return is not correctly represented by a single two-terminal thermal conductance.

through a larger cross-section increases passive thermal coupling. This geometry-only conductance screen is archived from the pre-V3.4 design assessment; the source-face, three-terminal return multiport is now included in V3.4/V3.5 and documented earlier. Its assumed interface and bulk properties, rather than the electrical path topology, now dominate outstanding thermal uncertainty.

12 Geometric fidelity and remaining discretization risk

The first full R3.8 TetGen volume candidate contained 1,043,314 tetrahedra over 430 material regions, with all 320,522 registered facets and 378 finite-contact witnesses retained. R3.9 applied 33,195 conservative 3-to-2 internal connectivity operations and reduced the mesh to 1,010,119 tetrahedra with unchanged original surface facets and regional volumes. R3.9 retains 241,754 tetrahedra with mean-ratio quality below 0.05 (23.93 percent)

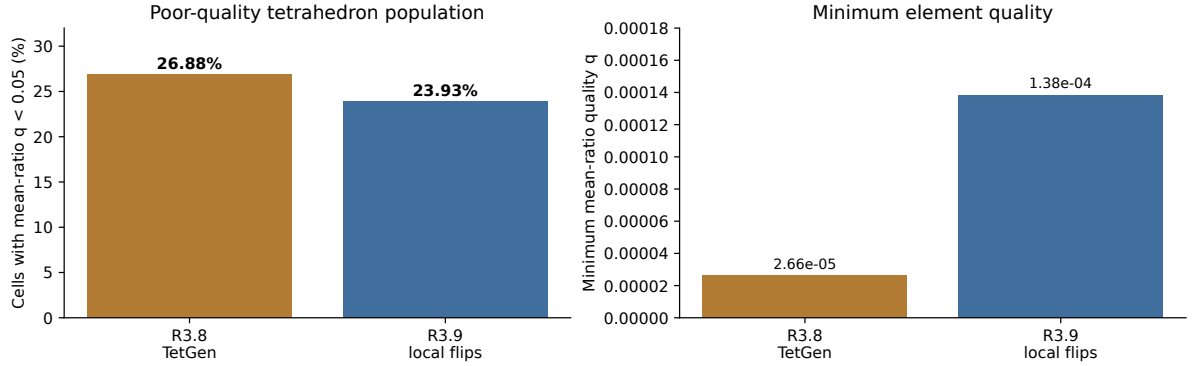


Figure 19: Independent archived global mesh-quality measurements. The 3-to-2 flips improved the poor-cell population, but the minimum quality and remaining weak cells prevent assuming mesh-converged coupled FEM.

and a minimum of 1.38495×10^{-4}. Thus no global coupled 430-domain electrical and thermal PDE is released. Historical full-solid electrical Gate B is FAIL/STOP; the first source N/P pair had eight raw gradient-current failures; the prospective P2 and global C/D computations are not qualified. Good energy closure in a separate hybrid model does not erase these original raw gates.

13 Thermoelectric material behavior and heat pumping

Each of the 88 axial P/N legs uses a signed 0.4 A series current and 32 longitudinal intervals. With z directed hot-to-cold, $S(T)$ Seebeck coefficient, $\rho(T)$ resistivity, $k(T)$ conductivity, and signed current density J, the constitutive conservation model is

$$q = S(T)TJ - k(T)\frac{dT}{dz}, \qquad \frac{dq}{dz} = J\left[\rho(T)J + S(T)\frac{dT}{dz}\right], \qquad \frac{dV}{dz} = \rho(T)J + S(T)\frac{dT}{dz}.$$

Midpoint constitutive evaluation and damped Newton iterations enforce thermal and electrical conservation at local temperatures, including the boundary identity $Q_h - Q_c = IV_{\text{leg}}$. The material source is a 14-knot figure digitization of research P-HE and N-HE-1 specimens over 40–300 K. Above 300 K the code uses an *unmeasured* one-quarter final-segment slope continuation; this is not a production material characterization.

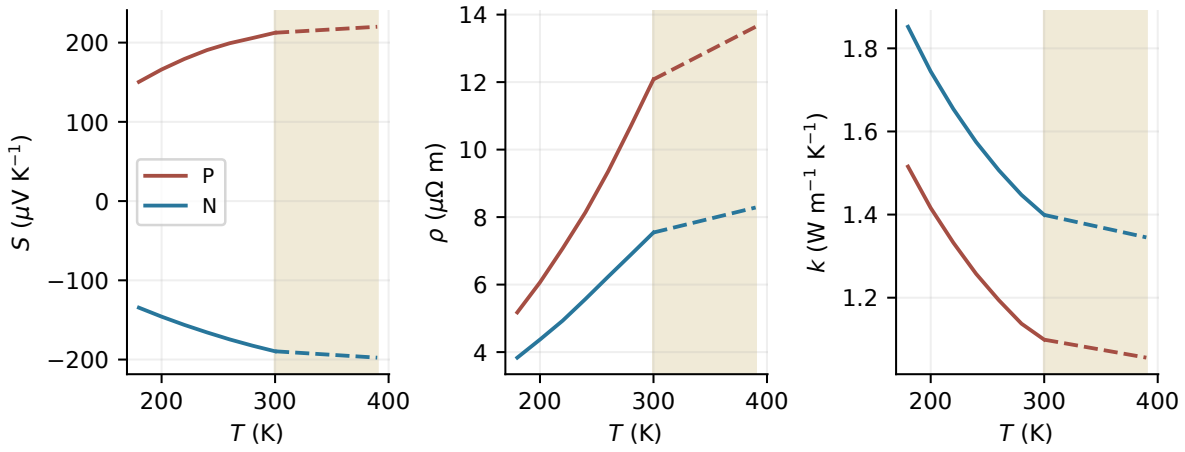


Figure 20: Source-derived P/N research analog properties. The shaded/dashed extension beyond 300 K is explicitly assumed; 58 hot-stage leg contacts exceed the research marker range.

14 Heat spreading through ceramic substrates

Each ceramic independently solves

$$-\nabla \cdot (k_c \nabla T) = 0, \qquad K_{ab} = \sum_e \int_{\Omega_e} k_c \nabla N_a \cdot \nabla N_b \, d\Omega, \quad b_a = \sum_f \int_{A_f} N_a q_f \, dA,$$

with $k_c = 30\,\mathrm{W/(m\,K)}$ a selected constant-property alumina analogue. Original face-specific P/N leg heat and assumed joint/cavity contributions are distributed over native TRI3 pad faces; source-local pad means feed back into leg end temperatures. Four archived source meshes contain 4,936 / 4,085 / 1,986 / 784 TET4 elements, totaling 11,791 tetrahedra.

14.1 Preserved spatial evidence: original V3.3 and current V3.5

The original source-ceramic temperature field from the earlier V3.3 reference is retained below as a *historical baseline*. Its thermal state omits the later copper-interstage multiport and independently located TEC lead boundary conditions; it must not be mistaken for a field from the present operating-point calculation.

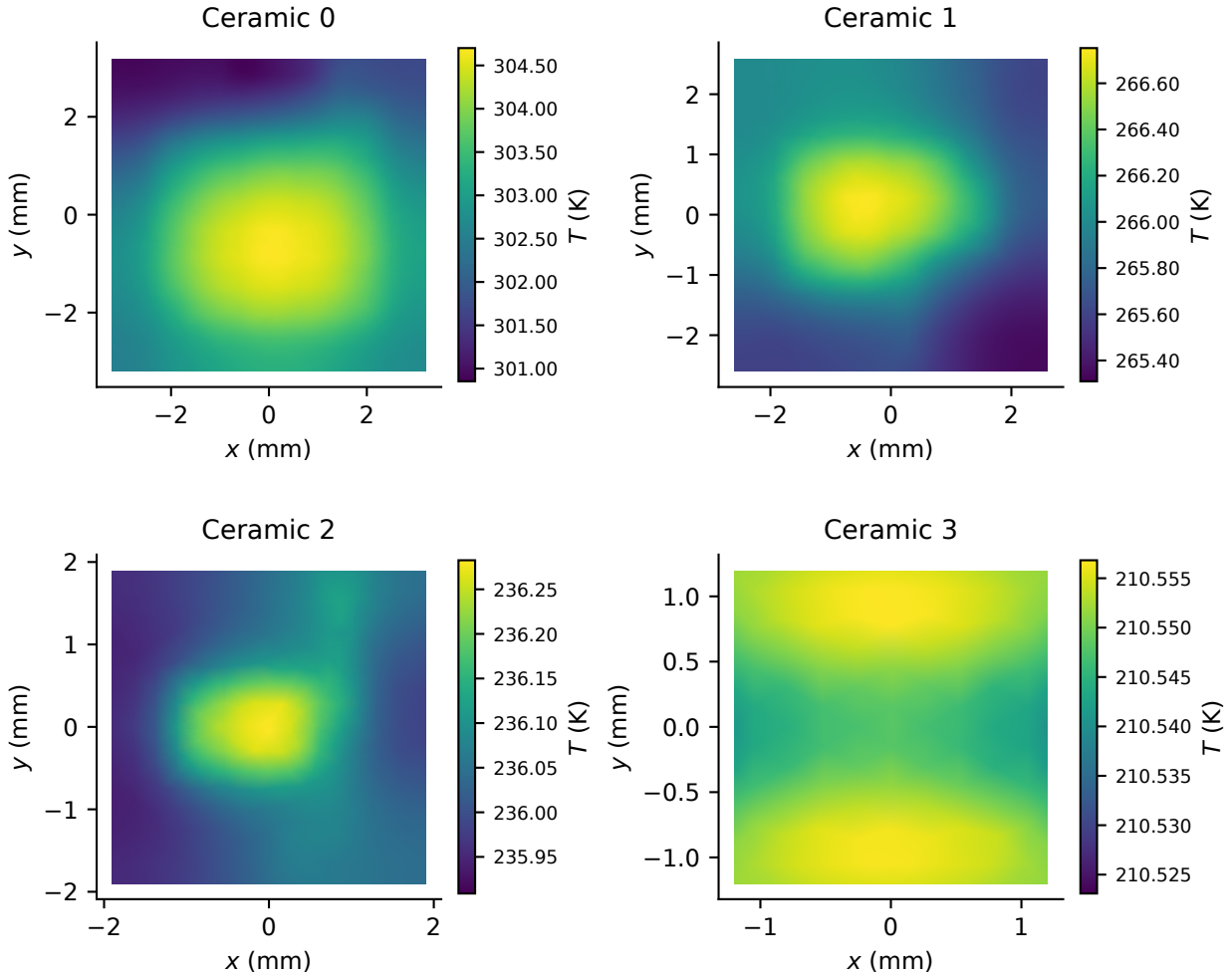


Figure 21: Preserved original V3.3 four-ceramic spatial FEM fields and source tetrahedral surface triangulations, calculated for the older reference without the V3.4 formed-copper heat network or V3.5 corrected TEC lead-pad heat. This is the original evidence, not a replacement or a claim of validated device temperature.

14.2 Current V3.5 source-ceramic temperature field

The revised V3.5 temperature plots use the *same original ceramic discretization* but a different, more complete interstage copper/terminal-wire heat budget. These two figures intentionally coexist: the old field is preserved, and the updated field is appended with its own physical assumptions. The four fields share their temperature gauge

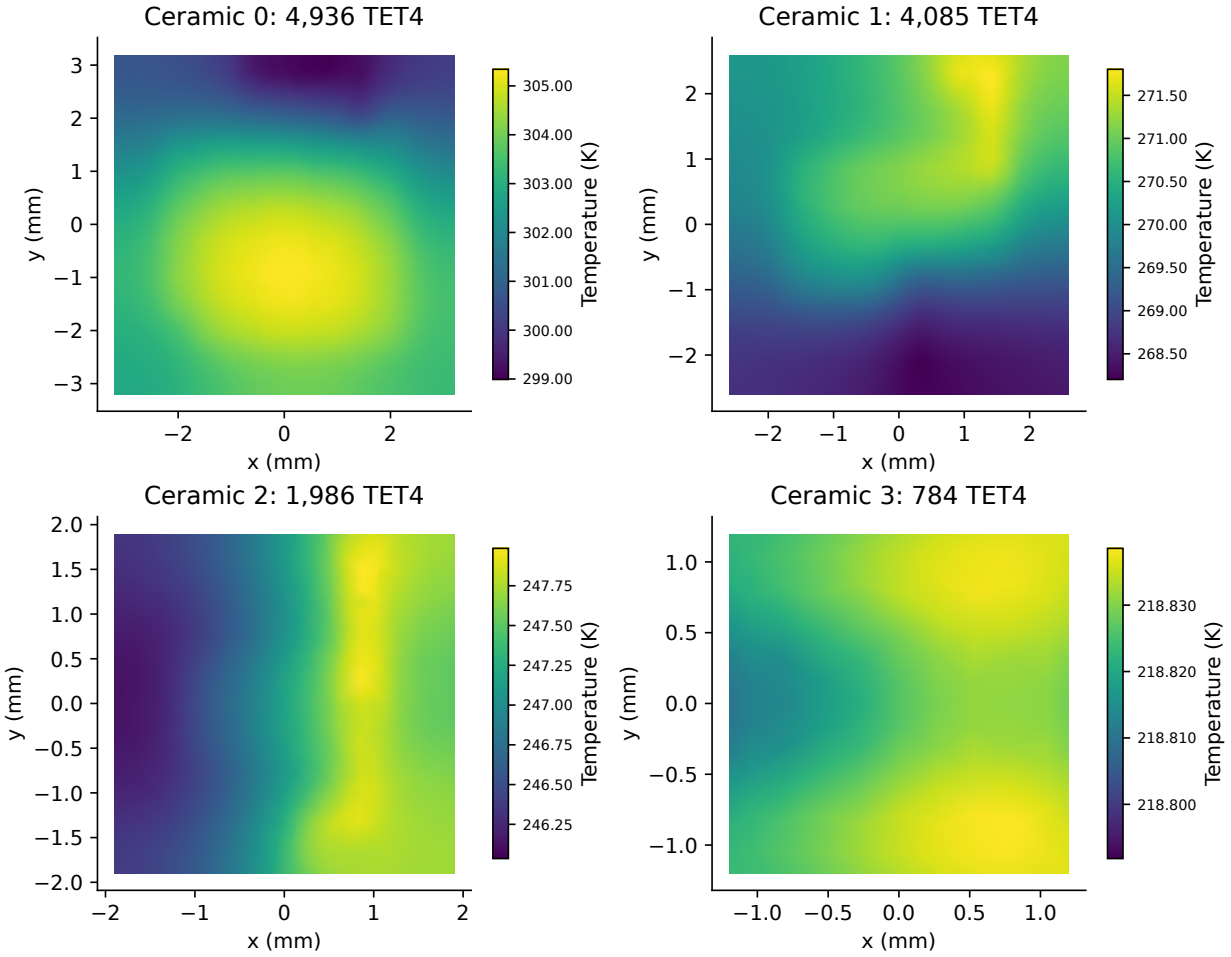


Figure 22: Accepted current V3.5 operating-point temperature on the original frozen 11,791 source-ceramic TET4 elements. The original upper-surface TRI3 boundary is colored by the saved spatial FEM nodal solution at 0.400 A, 300 K, 2 mW detector load, 100 Pa krypton and nominal Cu contacts. These are actual solved ceramic temperature fields; there is no claim of full 430-domain bulk-metal electrothermal FEM.

and stage heat exchange with the nonlocal reduced network. This is a single-mesh spatial result; no independent mesh-converged thermal accuracy bound is available.

15 Electrical contacts, solder joints and dissipative heating

The preserved source-solder model assumes 176 solder bodies of 35 micrometres thickness and 0.16 mm^2 area, with assumed SnSb resistivity $1.7 \times 10^{-7}\ \Omega\,\mathrm{m}$ and 352 assumed metal-interface specific resistances $r_c'' = 10^{-11}\ \Omega\,\mathrm{m}^2$. For each joint the resistance is

$$R_j = \rho_{\mathrm{SnSb}} \frac{L_j}{A_j} + 2\frac{r_c''}{A_j}, \quad R_{\mathrm{all}} = 0.028545\ \Omega, \quad I^2 R_{\mathrm{all}} = 4.56720\ \mathrm{mW}.$$

The implementation places 25.950 micowatts per source joint into its neighboring ceramic pad. This assumes the entire joint Joule heat reaches a ceramic and omits finite thermal-contact drops, interfacial thermopower and nonlocal copper-network heating. The four ceramic applied-joint terms, in mW, are [1.50510, 2.07600, 0.77850, 0.20760]. External TEC supply Cu wires and detector Au bonds are separate electrical/thermal links. *The formed tabs/return are accounted for by the separate V3.4/V3.5 reduced copper network described earlier, not by the solder-only term above.*

16 Package parasitic heat loads and thermal rejection

The reference forward case imposes a 300 K cap/window/ambient thermal reservoir, a 100 Pa krypton environment, an effective 3.0 mm gas gap, accommodation 0.8, and nominal gas conductivity knots $k_{\mathrm{Kr}}(200, 300, 400\,\mathrm{K}) = (0.0064, 0.0095, 0.0122)\ \mathrm{W/(m\,K)}$. Gray radiation uses assumed emissivities $\epsilon_{\mathrm{cap}} = 0.20$, $\epsilon_{\mathrm{ceramic}} = 0.80$, and $\epsilon_{\mathrm{window}} = 0.10$, with selected effective view factors. The source effective exposed surface areas are [12.60, 8.68, 4.76, 1.00] mm^2. The generic exchange terms are

$$Q_{\mathrm{gas},i} = G_{\mathrm{eff,i}}(p,T)(T_\infty - T_i), \qquad Q_{\mathrm{rad},i} = \epsilon_{\mathrm{eff},i}\sigma A_i F_i (T_\infty^4 - T_i^4).$$

These are *effective engineering closures*, not cavity fluid dynamics, true free-molecular transport, exact exposed-face loads, spectral window modeling or full radiosity.

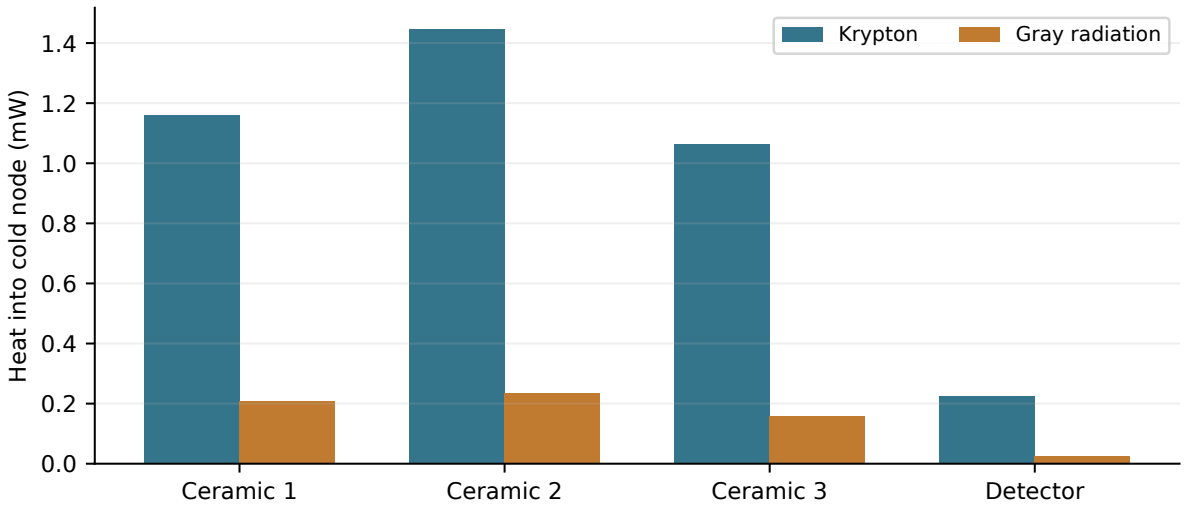


Figure 23: Archived V3.3 cavity contributions: total gas 3.892553 mW, gray radiation 0.624076 mW, combined heat into cold components 4.516629 mW. The reservoir carries an equal and opposite signed load.

Cold destination	Kr gas (mW)	Gray rad. (mW)	Total (mW)
Ceramic C1	1.15901	0.20874	1.36775
Ceramic C2	1.44533	0.23389	1.67923
Ceramic C3	1.06469	0.15770	1.22239
Detector	0.22352	0.02375	0.24726
TOTAL	3.89255	0.62408	4.51663

An effective header/TIM/radiator resistance of approximately 4.8334 K/W carries heat to the warm sink. Neither mount pressure nor material lot is an experimental datum. The cavity load is redistributed to native *upper metal-pad* sources in the current model; thermal gradients on exposed bare ceramic faces are not thereby demonstrated.

17 Predicted detector temperature under enclosure loading

The following **historical baseline** V3.2 versus V3.3 comparison (prior to adding V3.4 interstage copper) uses the same 88-leg source geometry, 11,791 ceramic tetrahedra, 0.4 A input, 300 K reservoir, 2 mW detector load and assumed material/contact parameters. The cavity raises die temperature 0.637233 K, while the nonlinear terminal voltage changes slightly due to changed operating gradients.

Metric	V3.2 (cavity off)	V3.3 (cavity on)
Detector temperature (K)	209.987981	210.625214
Terminal voltage (V)	1.524071	1.522462
Electrical input (W)	0.609628	0.608985
Bath rejection (W)	0.590323	0.594044
Pressure (Pa)	0 (cavity off)	100 (Kr)

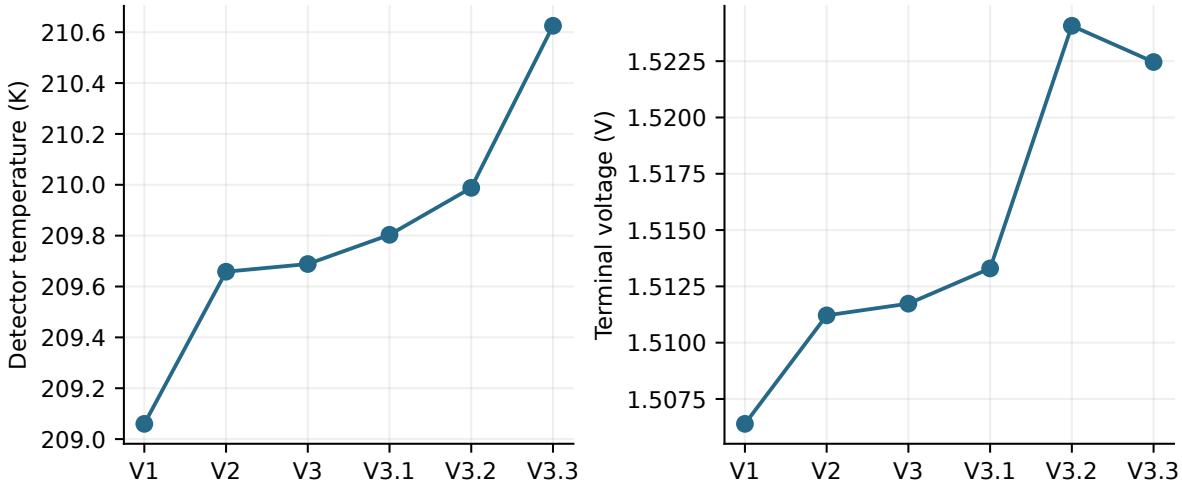


Figure 24: Recorded V1 to V3.3 complete reference-model milestones, all evaluated at the nominal current. These points describe changing model fidelity, not successive mesh convergence or independent measurements.

18 Historical V3.3 stage temperatures, heat flow, and electrical input

Stage	Legs	Cold flux (mW)	Hot flux (mW)	V	Joule (mW)
1	58	179.9602	614.7396	1.086948	262.732
2	22	46.1636	176.5165	0.325882	81.390
3	8	5.1576	43.7058	0.096371	24.642

Coupled node	Temperature (K)
Stage0 Hot	303.65655
Stage0 Cold	265.90329
Stage1 Hot	266.34287
Stage1 Cold	236.00259
Stage2 Hot	236.15716
Stage2 Cold	210.52765
Carrier	210.55066
Die	210.62521

The external terminal voltage includes additional lead and joint contributions beyond summed P/N voltage. Stage energies must account for signed electrical work, source losses, detector heat, cavity sources and external sink. Directly adding cavity milliwatts to input electrical power without solving the changed stage temperatures would be incorrect.

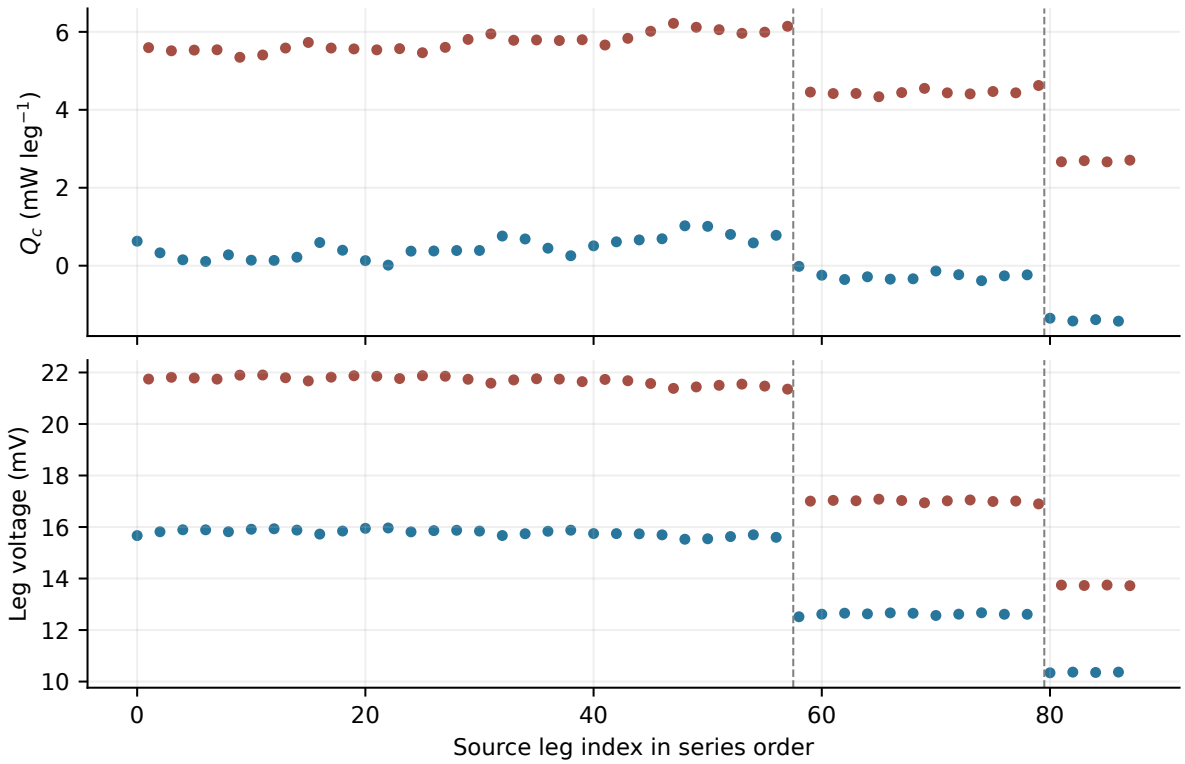


Figure 25: Real saved individual-leg cold-side heat and terminal-voltage contributions. Stage transitions occur after source leg indices 57 and 79. Source material laws and local ceramic pad temperatures determine leg-specific behavior.

19 Historical V3.3 energy conservation and numerical credibility

For the **earlier V3.3 baseline only**, the solver used the previous V3.2 solution as a warm start and iterated five spatial leg/pad updates, stopping at a maximum pad temperature change 1.65345×10^{-5} K below its registered 2×10^{-5} K criterion. That number is a *numerical stopping threshold*, not a temperature accuracy bound.

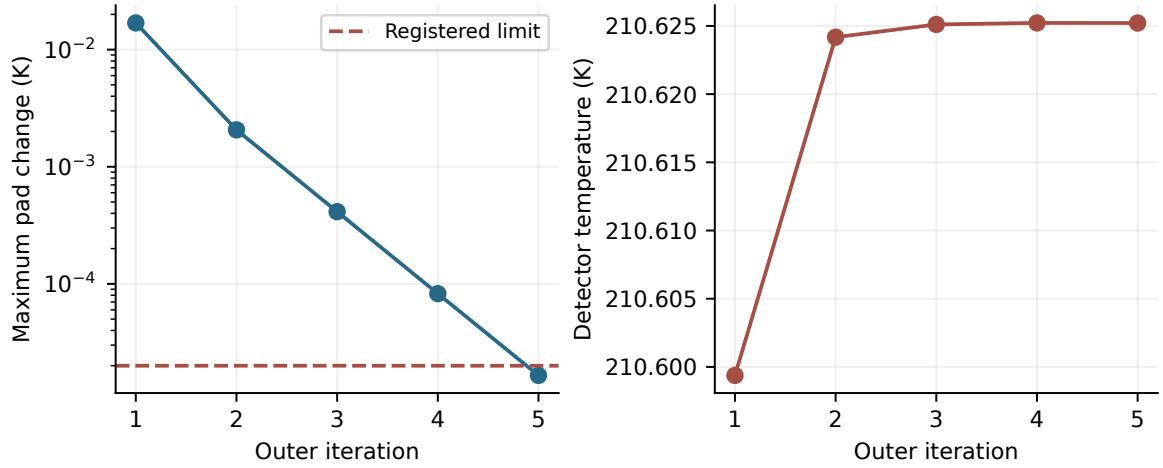


Figure 26: Actual saved spatial fixed-point iteration changes and detector temperature history. The last update meets the registered coarse convergence threshold; no mesh refinement or material uncertainty bound is inferred.

Iteration	Die T (K)	Max pad update (K)	Local flux mismatch (W)
1	210.599377	1.690e-02	3.914e-15
2	210.624173	2.066e-03	4.580e-16
3	210.625110	4.135e-04	1.478e-15
4	210.625220	8.254e-05	2.928e-15
5	210.625214	1.653e-05	5.503e-15

Metric	Absolute residual	Unit
Global energy	8.8674e-09	W
Worst stage balance	4.9061e-09	W
Ceramic pad heat	5.5025e-15	W
Worst individual leg	1.2364e-13	W
Final pad temperature change	1.6535e-05	K

An independent read-only field/load audit reconstructs the saved energy terms without rerunning the original PDE and finds a finite 2.535×10^{-8} W pad/leg update discrepancy. Numerical closure validates the *declared equations and discretization bookkeeping* at this operating point, not prediction fidelity of the as-built cooler.

20 Device engineering implications and next experiment

The physical cooler is now represented by a functioning complete thermal/electrical model including a first conservative three-branch copper network tied to original ceramic pads. The present V3.5 corrected-power-lead reference gives 218.900 K (versus V3.4 218.833 K and earlier V3.3 210.625 K); these are conditional modeled values, not manufactured device temperatures. It is *not* necessary to restart cooler CAD development. However, full three-dimensional copper/metal fields and actual thermal-contact characterization are not yet available. The current and detector-heat-load operating envelope now exists as the accepted V3.6 extension above. The next practical milestone is to couple bounded HgCdTe photoconductor resistance and bias/optical dissipation to that solved thermal environment, while quantifying uncertainty in the *already integrated* multiport: copper conductivity/resistivity versus temperature, joint thermal/electrical impedances, foil thickness and metal-film heat spreading. Use targeted source-informed screening to select conditions that merit expensive full-system recomputation. Full 3D copper FEM may be justified if compact validated metal substructures cannot bound current crowding or nonlocal heat flow. The older 38-page Revision 3 is preserved because of its unusually useful FEM methodology and publication-level plots, *not* because it describes the current metal layout. The 43-page interim extension is historical, not the latest design reference.

21 Model provenance and numerical evidence hierarchy

Artifact	Subject	Verified status
Historical Rev3	Earlier 216-solid 28/12/4 cooler	Detailed field method; raw-gradient release failed
Replacement V3 R1	824 physical STEP bodies, formed tabs and joints	Conditional CAD design acceptance
R3.9 geometry	430 material regions, 2,440 CAD faces	Geometric mesh audit; conditioning unqualified
V3.3 thermal baseline	88 nonlinear legs and four source 3D ceramic fields	Accepted reference without formed Cu network
V3.4 coupled source Cu	Source ceramics plus formed Cu multiport	Archived accepted reference; combined power-lead heat
V3.5 separated TEC leads	Source-local pin 1/9 wire heat and joint sensitivity	Three accepted conditional results; original 3D metal PDE still unqualified

The cooler envelope is informed by TEC Microsystems MC04 family reference geometry; the package by a separate VIGO nine-pin TO-66 drawing. They are not identified as a commercially fabricated composite. The 29/11/4 stage split, material grades, solder properties and fabricated metal details are source-inspired or chosen synthetic hypotheses, *not* proprietary OEM internals.

22 Physical assumptions and reproducibility specification

An independent implementation should use the four 0.5 mm-thick 6.4/5.2/3.8/2.4 mm ceramics, the Appendix B 88 leg centers and polarities, the 29/11/4-couple electrical series order and the source-selected 0.4 by 0.4 by 0.5 mm P/N geometry. Apply all six 14-knot P/N research-material curves in Appendix A; interpolate within the experimental range and explicitly impose the unmeasured quarter-final-segment-slope continuation above 300 K. Solve 88 individually signed 32-interval nonlinear axial thermoelectric legs and four source-geometry P1 ceramic steady conduction systems using local native TRI3 heat-face integrals. Set the declared assumed ceramic conductivity to 30 W/(m K), add the nominal 176 SnSb joints and 352 assumed electrical contacts, Cu and Au external leads, 2 mW detector heat, the stated finite mounting resistance and the effective 100 Pa Kr/gray radiation closure. For V3.5, include distinct pin1/9 supply lead heat on faces 8 and 36, then include the three finite-contact Cu-foil connections from native faces 38/75, 79/105 and 36/74/104 and iterate to a maximum local pad change below 0.0000003 K and independently check all signed stage and total heat balances. The archived Appendix C face centroids, areas and flux totals permit a comparable heat-integral replication, not byte-exact nodal temperatures. Bitwise reproduction requires original tetrahedral connectivity, exact face polygon vertices, quadrature and solver conventions. No unpublished real-world detector measurements are claimed.

23 Scientific and engineering references

1. TEC Microsystems, three-stage 3MC04-044-05 cooler family datasheet (2019), external envelope and thermoelectric reference. Its hidden internal routing is unknown.

2. VIGO Photonics, 3TE TO-66 nine-pin housing reference, Rev. 9 (2024). The cooler and housing are independent reference products.

3. J. Zhang et al., low-temperature bismuth telluride thermoelectric measurement and error-analysis study, *Chinese Physics Letters* 42, 056304 (2025). DOI 10.1088/0256-307X/42/5/056304. Source P-HE and N-HE-1 specimens are not the actual installed module.

4. M. L. Huber, gas thermal conductivity tabulations (NIST/CRC, 2019): krypton material reference, not experimentally measured gap conduction.

5. TEC Microsystems and Kryotherm multistage construction drawings and photographs; Coherent mounting guidelines. These establish process families, not qualification of the selected synthetic formed-tab design.

A Complete 14-knot P and N transport source tables

The following 84 values comprise six figure-digitized research-specimen curves. They do not establish actual commercial material properties. The assumed extension beyond 300 K is

$$f(T) = f(300) + 0.25\,(T - 300)\frac{f(300) - f(280)}{20}, \qquad f \in \{S, \rho, k\}.$$

T K	S_P μV/K	ρ_P $\mu\Omega$ m	k_P W/(m K)	S_N μV/K	ρ_N $\mu\Omega$ m	k_N W/(m K)
40	31.477086	1.226120	2.776424	-39.762194	0.837035	3.928920
60	44.991486	1.550968	2.488935	-53.616397	1.156199	3.246955
80	60.679339	1.955036	2.260290	-67.469437	1.522950	2.816874
100	77.996604	2.446265	2.072406	-81.936140	1.920207	2.536709
120	96.944423	3.004830	1.909416	-95.992742	2.354830	2.310364
140	115.755946	3.567364	1.764537	-109.847998	2.820018	2.129254
160	133.684757	4.343818	1.630978	-122.547039	3.319204	1.973635
180	150.458980	5.175731	1.515526	-134.499967	3.835302	1.851971
200	166.078757	6.067058	1.415926	-145.840430	4.371811	1.744414
220	179.321421	7.073284	1.332165	-156.231255	4.932113	1.653884
240	190.594863	8.138932	1.257469	-165.737958	5.570441	1.574639
260	199.219885	9.351143	1.194083	-174.568277	6.232621	1.506729
280	205.739245	10.694094	1.137484	-182.445326	6.884545	1.447332
300	212.462550	12.080611	1.099006	-189.575244	7.543277	1.399227

B All original 88 P/N leg centers

Each leg has 0.1600 mm^2 area and 0.5000 mm axial length; stage index 0 is the hottest. All coordinates refer to the original selected synthetic CAD x/y frame in mm.

No.	Stage	P/N	x (mm)	y (mm)	No.	Stage	P/N	x (mm)	y (mm)
0	0	N	-2.030000	-2.030000	44	0	N	-0.290000	0.870000
1	0	P	-1.450000	-2.030000	45	0	P	-0.870000	0.870000
2	0	N	-0.870000	-2.030000	46	0	N	-1.450000	0.870000
3	0	P	-0.290000	-2.030000	47	0	P	-2.030000	0.870000
4	0	N	0.290000	-2.030000	48	0	N	-2.030000	1.450000
5	0	P	0.870000	-2.030000	49	0	P	-1.450000	1.450000
6	0	N	1.450000	-2.030000	50	0	N	-0.870000	1.450000
7	0	P	2.030000	-2.030000	51	0	P	-0.290000	1.450000
8	0	N	2.030000	-1.450000	52	0	N	0.290000	1.450000
9	0	P	1.450000	-1.450000	53	0	P	0.870000	1.450000
10	0	N	0.870000	-1.450000	54	0	N	1.450000	1.450000
11	0	P	0.290000	-1.450000	55	0	P	2.030000	1.450000
12	0	N	-0.290000	-1.450000	56	0	N	2.030000	2.030000
13	0	P	-0.870000	-1.450000	57	0	P	1.450000	2.030000
14	0	N	-1.450000	-1.450000	58	1	N	1.450000	0.870000
15	0	P	-2.030000	-1.450000	59	1	P	0.870000	0.870000
16	0	N	-2.030000	-0.870000	60	1	N	0.290000	0.870000
17	0	P	-1.450000	-0.870000	61	1	P	-0.290000	0.870000
18	0	N	-0.870000	-0.870000	62	1	N	-0.870000	0.870000
19	0	P	-0.290000	-0.870000	63	1	P	-1.450000	0.870000
20	0	N	0.290000	-0.870000	64	1	N	-1.450000	0.290000
21	0	P	0.870000	-0.870000	65	1	P	-0.870000	0.290000
22	0	N	1.450000	-0.870000	66	1	N	-0.290000	0.290000
23	0	P	2.030000	-0.870000	67	1	P	0.290000	0.290000
24	0	N	2.030000	-0.290000	68	1	N	0.870000	0.290000
25	0	P	1.450000	-0.290000	69	1	P	1.450000	0.290000
26	0	N	0.870000	-0.290000	70	1	N	1.450000	-0.290000
27	0	P	0.290000	-0.290000	71	1	P	0.870000	-0.290000
28	0	N	-0.290000	-0.290000	72	1	N	0.290000	-0.290000
29	0	P	-0.870000	-0.290000	73	1	P	-0.290000	-0.290000
30	0	N	-1.450000	-0.290000	74	1	N	-0.870000	-0.290000
31	0	P	-2.030000	-0.290000	75	1	P	-1.450000	-0.290000
32	0	N	-2.030000	0.290000	76	1	N	-1.450000	-0.870000
33	0	P	-1.450000	0.290000	77	1	P	-0.870000	-0.870000
34	0	N	-0.870000	0.290000	78	1	N	-0.290000	-0.870000
35	0	P	-0.290000	0.290000	79	1	P	0.290000	-0.870000
36	0	N	0.290000	0.290000	80	2	N	0.870000	-0.290000
37	0	P	0.870000	0.290000	81	2	P	0.290000	-0.290000
38	0	N	1.450000	0.290000	82	2	N	-0.290000	-0.290000
39	0	P	2.030000	0.290000	83	2	P	-0.870000	-0.290000
40	0	N	2.030000	0.870000	84	2	N	-0.870000	0.290000
41	0	P	1.450000	0.870000	85	2	P	-0.290000	0.290000
42	0	N	0.870000	0.870000	86	2	N	0.290000	0.290000
43	0	P	0.290000	0.870000	87	2	P	0.870000	0.290000

C Native ceramic-face geometric and solved heat ledger

For each original ceramic and native thermal face, the table lists source face ID, centroid (mm), original CAD area (mm^2) and the signed integrated solved heat *into* that ceramic (W). Contributions include local P/N transport, provisional joint heating, finite sink, detector and effective cavity loads when applicable. Exact face polygons are not fully specified by centroids and areas.

Cer.	Face ID	x	y	z	Area mm^2	Heat in (W)
0	6	0.000000	0.000000	1.584250	37.21000000	-5.9404375264e-01
0	8	-2.030000	-2.415000	2.084250	0.58100000	+6.9212449542e-03
0	9	-1.160000	-2.030000	2.084250	0.54000000	+2.1001118011e-02
0	10	-0.000000	-2.030000	2.084250	0.54000000	+2.0799385661e-02
0	11	-2.030000	-1.160000	2.084250	0.54000000	+2.1333430585e-02
0	12	-1.160000	-1.450000	2.084250	0.54000000	+2.0925106301e-02
0	13	-1.160000	-0.870000	2.084250	0.54000000	+2.1098181465e-02
0	14	-2.030000	-0.000000	2.084250	0.54000000	+2.1661444954e-02
0	15	-1.160000	-0.290000	2.084250	0.54000000	+2.1280527762e-02
0	16	-0.000000	-1.450000	2.084250	0.54000000	+2.0726535211e-02
0	17	0.000000	-0.870000	2.084250	0.54000000	+2.0872339885e-02
0	18	-0.000000	-0.290000	2.084250	0.54000000	+2.1135046258e-02
0	19	1.160000	-2.030000	2.084250	0.54000000	+2.0762274895e-02
0	20	2.030000	-1.740000	2.084250	0.54000000	+2.0894962316e-02
0	21	1.160000	-1.450000	2.084250	0.54000000	+2.0664089207e-02
0	22	1.160000	-0.870000	2.084250	0.54000000	+2.0730610371e-02
0	23	1.160000	-0.290000	2.084250	0.54000000	+2.0987932670e-02
0	24	2.030000	-0.580000	2.084250	0.54000000	+2.1025501033e-02
0	25	-1.160000	0.290000	2.084250	0.54000000	+2.1501510390e-02
0	26	-2.030000	1.160000	2.084250	0.54000000	+2.2056807823e-02
0	27	-1.160000	0.870000	2.084250	0.54000000	+2.1665224003e-02
0	28	-1.160000	1.450000	2.084250	0.54000000	+2.1972015528e-02
0	29	0.000000	0.290000	2.084250	0.54000000	+2.1329922044e-02
0	30	-0.000000	0.870000	2.084250	0.54000000	+2.1513476612e-02
0	31	0.000000	1.450000	2.084250	0.54000000	+2.1762448001e-02
0	32	1.160000	0.290000	2.084250	0.54000000	+2.1132384800e-02
0	33	2.030000	0.580000	2.084250	0.54000000	+2.1315494607e-02
0	34	1.160000	0.870000	2.084250	0.54000000	+2.1316483722e-02
0	35	1.160000	1.450000	2.084250	0.54000000	+2.1497446937e-02
0	36	0.088570	2.858995	2.084250	0.67225000	-2.2200926469e-02
0	37	2.030000	1.740000	2.084250	0.54000000	+2.1652376653e-02
0	38	1.450000	2.379622	2.084250	0.51850000	+1.4709356454e-02
1	44	-1.740000	-2.030000	2.684250	0.54000000	-6.1717204457e-03
1	45	-0.580000	-2.030000	2.684250	0.54000000	-5.7923394592e-03
1	46	0.580000	-2.030000	2.684250	0.54000000	-5.6313481447e-03
1	47	-1.740000	-1.450000	2.684250	0.54000000	-5.8945158835e-03
1	48	-1.740000	-0.870000	2.684250	0.54000000	-6.1272498337e-03
1	49	-0.580000	-1.450000	2.684250	0.54000000	-5.6677985879e-03
1	50	-0.580000	-0.870000	2.684250	0.54000000	-5.9073535857e-03
1	51	-1.740000	-0.290000	2.684250	0.54000000	-6.2861601984e-03
1	52	-1.740000	0.290000	2.684250	0.54000000	-6.4929448577e-03
1	53	-0.580000	-0.290000	2.684250	0.54000000	-6.1441766581e-03
1	54	0.580000	-1.450000	2.684250	0.54000000	-5.4949366062e-03
1	55	0.580000	-0.870000	2.684250	0.54000000	-5.6165637076e-03
1	56	0.580000	-0.290000	2.684250	0.54000000	-5.9284010470e-03
1	57	1.740000	-2.030000	2.684250	0.54000000	-5.5964450493e-03
1	58	1.740000	-1.450000	2.684250	0.54000000	-5.5750930796e-03
1	59	1.740000	-0.870000	2.684250	0.54000000	-5.5316729855e-03
1	60	1.740000	-0.290000	2.684250	0.54000000	-5.7862099653e-03
1	61	-0.580000	0.290000	2.684250	0.54000000	-6.4257386885e-03
1	62	-1.740000	0.870000	2.684250	0.54000000	-6.8576261766e-03
1	63	-1.740000	1.450000	2.684250	0.54000000	-7.0890067759e-03
1	64	-0.580000	0.870000	2.684250	0.54000000	-6.6249732312e-03
1	65	-0.580000	1.450000	2.684250	0.54000000	-7.0137372522e-03
1	66	0.580000	0.290000	2.684250	0.54000000	-6.1746390641e-03
1	67	0.580000	0.870000	2.684250	0.54000000	-6.3948651910e-03
1	68	0.580000	1.450000	2.684250	0.54000000	-6.7126784447e-03
1	69	1.740000	0.290000	2.684250	0.54000000	-6.0016216656e-03
1	70	1.740000	0.870000	2.684250	0.54000000	-6.1198269805e-03
1	71	1.740000	1.450000	2.684250	0.54000000	-6.5247521871e-03
1	72	1.740000	2.030000	2.684250	0.54000000	-6.8707555753e-03
1	74	0.870000	2.300000	3.184250	0.12000000	+2.3941388903e-05
1	75	1.450000	1.470862	3.184250	0.67900000	+5.1504788556e-03
1	76	-1.450000	-0.580000	3.184250	0.54000000	+1.6210670281e-02
1	77	-0.580000	-0.870000	3.184250	0.54000000	+1.6207954022e-02
1	78	-0.580000	-0.290000	3.184250	0.54000000	+1.6071664242e-02
1	79	0.290000	-1.435162	3.184250	0.65650000	+1.1539642240e-02
1	80	0.580000	-0.290000	3.184250	0.54000000	+1.6217492734e-02
1	81	1.450000	0.000000	3.184250	0.54000000	+1.6376311378e-02
1	82	-1.450000	0.580000	3.184250	0.54000000	+1.6153119097e-02
1	83	-0.580000	0.290000	3.184250	0.54000000	+1.6048683270e-02
1	84	-0.580000	0.870000	3.184250	0.54000000	+1.6098735692e-02
1	85	0.580000	0.290000	3.184250	0.54000000	+1.6138044094e-02
1	86	0.580000	0.870000	3.184250	0.54000000	+1.6218414034e-02

Cer.	Face ID	x	y	z	Area	Heat (W)
2	92	-1.160000	-0.870000	3.784250	0.54000000	-4.1229763184e-03
2	93	-1.160000	-0.290000	3.784250	0.54000000	-4.0325885849e-03
2	94	0.000000	-0.870000	3.784250	0.54000000	-4.3372223253e-03
2	95	-0.000000	-0.290000	3.784250	0.54000000	-4.1255544021e-03
2	96	1.160000	-0.290000	3.784250	0.54000000	-4.2477066490e-03
2	97	-1.160000	0.290000	3.784250	0.54000000	-3.9991911145e-03
2	98	-1.160000	0.870000	3.784250	0.54000000	-4.0132490723e-03
2	99	0.000000	0.290000	3.784250	0.54000000	-4.0452781812e-03
2	100	1.160000	0.290000	3.784250	0.54000000	-4.1617896577e-03
2	101	-0.000000	0.870000	3.784250	0.54000000	-4.1215621922e-03
2	102	1.160000	0.870000	3.784250	0.54000000	-4.3855401752e-03
2	104	0.870000	0.829062	4.284250	0.64000000	+8.5720985048e-03
2	105	0.692720	-0.929413	4.284250	0.81350000	+3.2580848072e-03
2	106	-0.000000	-0.290000	4.284250	0.54000000	+1.1238504436e-02
2	107	-0.870000	0.000000	4.284250	0.54000000	+1.1290111066e-02
2	108	0.000000	0.290000	4.284250	0.54000000	+1.1233859859e-02
3	114	0.580000	0.290000	4.884250	0.54000000	-1.2327805666e-03
3	115	-0.580000	-0.290000	4.884250	0.54000000	-1.2229792254e-03
3	116	0.580000	-0.290000	4.884250	0.54000000	-1.2662612022e-03
3	117	-0.580000	0.290000	4.884250	0.54000000	-1.2279447029e-03
3	119	-0.000000	-0.000000	5.384250	4.41000000	+4.9499656971e-03

D Verification boundaries and remaining work

Technical claim	Actual status	Appropriate requirement for improvement
824-solid replacement V3 CAD	Conditional physical-design PASS	Measured forming, solder, current and reliability capability
Native pin-to-pin topology	88/88 legs PASS	Finite real current density, resistance and contact temperature fields
Global 430-domain R3.9 mesh	Geometric PASS	Improve worst elements and prove solver conditioning/convergence
Original full-solid electrical Gate B	Historical FAIL/STOP	Qualify original raw-gradient tests on a viable mesh
Full 430-domain electrothermal PDE	NOT RUN	Integrated metal/contact/leg/ceramic field solution
V3.5 Cu and source-lead hybrid at 0.4 A	Conditional numerical PASS	Solder/foil interface and plated metal-sheet laws, then current/load response
Rarefied gas and radiation	Effective load model	Native exposed-surface gas/gray/spectral physics
Real thermoelectric grades	Unidentified	Bounded synthetic studies or actual module certificates
HgCdTe photodetection	No carrier/optical PDE	Detector noise/spectral/bias physics if needed
As-built product prediction	Unvalidated	Device measurements; do not claim manufacturer equivalence

Publication policy. The original 38-page Revision 3 remains a sealed separate historical methodology reference. Its later 43-page interim source-mesh integration is likewise archived. This document describes the newer replacement CAD and the most recent completed running hybrid with the unmodeled internal copper paths prominently disclosed. All figures and numerical tables derive from existing saved evidence, and no new physics calculation was performed for this publication.